

Structural Intelligence, Mathematical Research Landscape, and Research-Ecosystem Scale

An Integrated Long-Form Preprint on the Diffraction Profile, Mathematical Specialization, and the Size of Mathematical Research Communities

Integrated thesis.

A finite-resource theory of expertise is proposed as a candidate explanation for a *T-shaped profile with side lobes*: one deep vertical specialization, a broad competence floor, and selectively elevated secondary regions created by structural transfer. The theory is then instantiated as a proposed architecture for mathematical specialization in an AI-rich environment, and the surrounding mathematical research ecosystem is mapped quantitatively by approximate community size. The purpose of the integrated document is not to declare a universally optimal career or a permanent AI-proof zone, but to make the theory, the proposed mathematical architecture, and the empirical scale of the relevant research communities explicit enough to be tested, attacked, and revised.

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September 2026

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1. Structural Intelligence and the Diffraction Profile of Human Expertise

1.1 Scope and Abstract of the Foundational Theory

This paper proposes the *Diffraction Profile*, a mathematical hypothesis about the allocation of finite cognitive resources across a structured space of skills. Traditional I-, T-, Pi-, and Comb-shaped descriptions are useful as organizational metaphors but treat expertise as a small number of discrete blocks. The present framework instead models expertise as a nonnegative continuous profile with a baseline level of broad competence and a dominant central specialization. A squared-sinc kernel, motivated by the shape of single-slit Fraunhofer diffraction, is used as a parsimonious parametric family because it produces one dominant maximum together with progressively weaker secondary maxima. These secondary lobes are interpreted not as literal physical waves in cognition, but as a formal representation of *structural resonance*: domains that are not adjacent in ordinary disciplinary distance may nevertheless receive enhanced competence when they share deep representational structure with the core domain.

The paper is deliberately narrower than a theory of general intelligence. It proposes a model of expertise allocation and transfer, makes explicit the assumptions required for the model, derives exact normalization on a finite cognitive domain, distinguishes within-family optimization from a global optimum, and states empirical predictions that could in principle falsify the framework. The resulting theory provides a mathematical language for a T-shaped or “T-with-side-lobes” style of intellectual development: one deep vertical, a general competence floor, and selected secondary regions of competence produced by transfer rather than by independent full specialization. A companion mathematical-landscape preprint applies this hypothesis to a proposed architecture for mathematical specialization, while a companion statistical preprint estimates the scale of the corresponding research ecosystems. The three documents are therefore best viewed as separate but convergent layers: theory, architecture, and ecosystem.

Keywords: expertise, cognitive resource allocation, transfer of learning, structural similarity, specialization, polymathy, T-shaped skills, mathematical modeling, diffraction analogy, human–AI collaboration

1.2 Introduction

The allocation of finite cognitive resources across multiple domains is a persistent problem in education, career design, expertise research, and human capital. An individual who attempts to specialize in everything generally cannot acquire maximal depth everywhere. An individual who specializes extremely narrowly may achieve exceptional performance in one domain while becoming fragile when tasks, technologies, or research questions move outside that domain. Between these extremes lie familiar descriptions such as I-shaped, T-shaped, Pi-shaped, and Comb-shaped expertise.

These typologies are useful precisely because they are easy to communicate. They are much less useful as mathematical models. They generally represent competence as a collection of discrete blocks: one deep field, several shallow fields, two deep fields, or several deep fields. What is missing is a continuous account of how depth can vary across a structured space of skills and, more importantly, how transfer between skills might depend on structural similarity rather than on ordinary disciplinary adjacency.

This paper introduces the *Diffraction Profile* as a hypothesis designed to address that gap. The proposal is intentionally modest in one sense and ambitious in another. It is modest because it does not claim to define intelligence in general, nor does it claim that human cognition literally behaves like an optical wave. It is ambitious because it attempts to turn a familiar intuition about expertise—deep specialization can create unexpected competence elsewhere—into a formally specified, resource-constrained model.

The central intuition is:

A person may allocate most cognitive investment to a single core domain while acquiring nontrivial competence in multiple other domains, and those secondary concentrations of competence may be better predicted by *structural resonance* than by surface similarity or disciplinary distance.

The optical analogy is useful because a single-slit diffraction pattern has a dominant central maximum accompanied by smaller secondary lobes. The mathematical shape therefore provides a convenient prototype for a cognitive profile with one primary specialization and non-monotonic secondary concentrations.

The purpose of this paper is not to argue that the squared-sinc function is uniquely correct. It is to establish a coherent, falsifiable framework in which the following ideas can be stated precisely:

- (i) cognitive expertise is treated as a finite resource rather than an unlimited stock;
- (ii) domains are embedded in a structured space in which distance can represent conceptual or representational similarity;

- (iii) a dominant specialization can coexist with a nonzero baseline of broad competence;
- (iv) secondary competence can occur away from the core when structural resonance is high;
- (v) the shape of the profile generates testable predictions about transfer;
- (vi) a T-shaped career with side lobes can be viewed as a special case of a continuous resource-allocation function.

Scope of the claim. The Diffraction Profile is proposed here as a mathematical model and hypothesis about *expertise allocation and structural transfer*. It is not presented as a complete theory of intelligence, consciousness, general reasoning ability, or human cognition. The term “intelligence” in the title refers to the architecture of expertise and transfer being modeled, not to a psychometric definition of general intelligence.

1.3 Position Relative to the Companion Preprints

The present manuscript is one of three separate preprints developed as parts of a single research program.

1.3.1 Layer I: Theory

This paper introduces the Diffraction Profile. Its object is the geometry of expertise: how a finite resource can be concentrated around a core while still producing a broad competence floor and structurally resonant secondary regions.

1.3.2 Layer II: Mathematical architecture

The companion preprint *AI-Era Mathematical Specialization: A Deep Audit of a T-Shaped Mathematical Architecture* applies the theoretical idea to mathematics. Its proposed vertical is

$$\text{Algebraic Topology} \longrightarrow \text{Homotopy Theory} \longrightarrow \text{Homological Algebra},$$

with lateral engines including Abstract Algebra, Measure Theory and Functional Analysis, Dynamical Systems and Ergodic Theory, and Category Theory. The companion paper deliberately treats the vertical as a cognitive stem rather than as a claim of permanent AI resistance.

1.3.3 Layer III: Ecosystem statistics

The companion preprint *Research Ecosystem Size in Mathematics* provides demographic context: approximate orders of magnitude for the research communities occupying the proposed mathematical landscape. It distinguishes database-defined populations from active research

populations and distinguishes direct bibliometric evidence from triangulated estimates and community-scale proxies.

The relationship can therefore be represented schematically as

$$\boxed{\text{theory of expertise} \longrightarrow \text{proposed mathematical architecture} \longrightarrow \text{research-ecosystem map.}}$$

The three preprints are intended to remain separate. The present paper should be read as the foundational model; the mathematical-landscape paper is an application; the statistical paper is empirical context for that application.

1.4 Conceptual Background

1.4.1 I-, T-, Pi-, and Comb-shaped expertise

T-shaped descriptions emphasize one deep area together with broad supporting competence. Pi-shaped descriptions emphasize two deep areas, while Comb-shaped descriptions emphasize several. These metaphors have proved useful in management, professional education, and engineering discussions [6, 5, 9, 7].

The present proposal does not reject those metaphors. Instead, it asks what they look like when competence is represented continuously.

An idealized T-shape can be interpreted as a profile with one dominant maximum and a broad low-level floor. A Pi-shape can be represented by two dominant maxima. A Comb shape can be represented by many maxima. The key question is then whether the location and height of secondary maxima can be modeled as consequences of structural similarity rather than as independently chosen specializations.

1.4.2 Cognitive constraints and deliberate practice

Cognitive load research emphasizes the limited capacity of working memory during difficult problem solving [2]. Deliberate-practice research emphasizes the long time scale over which high levels of expertise are acquired [1]. Neither literature by itself establishes a literal scalar budget of “cognitive energy”, and the present paper therefore treats such a budget as a *modeling assumption* rather than an experimentally established constant.

This distinction matters. The equation

$$\int S(x) dx = E$$

is not intended to assert that brains physically conserve a quantity called cognitive energy in the same manner as a closed physical system conserves mechanical energy. Rather, it encodes an

economic and temporal intuition: effort devoted to one region is effort that cannot simultaneously be spent at full intensity in arbitrarily many other regions.

1.4.3 Transfer and identical elements

Thorndike and Woodworth's work on transfer emphasized the importance of common elements between tasks [3]. Subsequent transfer research has continued to distinguish relatively near transfer from difficult far transfer [4, 8].

The Diffraction Profile builds on this general idea but changes the geometry. Instead of assuming that competence decreases smoothly with disciplinary distance, it hypothesizes that competence may be elevated again at specific locations where deep structures resonate. In this sense, a secondary lobe is a hypothesis about *non-monotonic transfer*.

1.4.4 What the model does not assume

The framework does not assume that every person has the same profile. It does not assume that every secondary lobe represents positive transfer. It does not assume that structurally similar domains are necessarily adjacent in institutional classification. It does not assume that one core specialization is always superior to multiple cores. It does not claim that the squared-sinc kernel is the only admissible kernel.

1.5 Modeling Assumptions

The formal model rests on a small set of explicit assumptions.

Definition 1.5.1 (Skill space). Let $(\mathcal{X}, d)$ be a measurable metric space whose points represent domains, tasks, or cognitive functions. The distance $d(x, y)$ is not required to be geographical or institutional. It is intended to measure a chosen notion of *structural distance*: the difference between the representations, invariants, operations, or problem-solving architectures required by the two domains.

Definition 1.5.2 (Expertise profile). An expertise profile is a nonnegative function

$$S : \mathcal{X} \rightarrow [0, \infty)$$

where $S(x)$ represents the depth, fluency, or usable competence of an individual at location x .

Definition 1.5.3 (Core domain). A core domain $x_0 \in \mathcal{X}$ is a location at which the expertise profile attains its principal maximum.

Definition 1.5.4 (Structural resonance). For a fixed core x_0 , structural resonance is a property that may cause competence to remain elevated, or to rise again, at locations whose structural distance from x_0 is not small in ordinary disciplinary terms.

Hypothesis 1.5.1 (Finite-resource hypothesis). For a fixed individual and a fixed effective planning horizon, the integrated amount of investment represented by the expertise profile is bounded by a finite budget E .

Hypothesis 1.5.2 (Structural-transfer hypothesis). Transfer from a core domain is not necessarily a monotone function of nominal distance. Competence can display secondary maxima at structurally resonant locations.

Hypothesis 1.5.3 (Generalist-floor hypothesis). Sustained broad development creates a nonzero competence floor B over the effective skill space relevant to the individual's work.

These are hypotheses or modeling assumptions, not established empirical laws.

1.6 From a One-Dimensional Profile to a Structured Skill Space

The original intuition behind the model is one-dimensional because it is useful to draw. Let $x \in [-D, D]$ denote a coordinate centered at the core $x_0 = 0$. This one-dimensional version should be interpreted as a reduced representation of a more complicated space.

The one-dimensional model is pedagogically valuable because it makes the central claim visually transparent. However, the intended application to mathematics is inherently networked. Mathematical domains form a graph of conceptual transfer: topology connects to geometry and dynamics; algebra connects to representation theory and algebraic geometry; analysis connects to probability, dynamics, and PDE; category theory connects many of these structures. The companion mathematical-landscape preprint makes this graph interpretation explicit.

Accordingly, this paper distinguishes two levels of the model:

- (a) a **reduced one-dimensional model** used to derive and visualize the diffraction profile;
- (b) a **general structured-space model** in which the kernel depends on a structural distance $d(x, x_0)$.

The one-dimensional model is therefore not claimed to be the final geometry of cognition. It is the simplest mathematically tractable instantiation of the theory.

1.7 The Resource Constraint

In the one-dimensional representation, let the effective cognitive spectrum be the finite interval $[-D, D]$. The resource constraint is

$$\int_{-D}^D S(x) dx = E, \tag{1.1}$$

where $E > 0$ is the total modeled resource available to the profile.

The baseline B must satisfy

$$0 < B < E/(2D), \quad (1.2)$$

so that some resource remains for specialization above the baseline.

This formulation is intentionally abstract. Depending on the application, E may represent a weighted combination of time, sustained attention, practice effort, educational exposure, or other scarce resources. The model does not claim that these quantities are physically identical.

A crucial correction to the strongest form of the original argument is therefore appropriate:

Interpretive correction. The integral constraint should be read as a normalized resource-allocation model, not as a literal biological conservation law. The model is useful if it predicts observed patterns of specialization and transfer; its usefulness does not depend on the existence of a measurable biological substance called “cognitive energy.”

1.8 The Diffraction Profile

The proposed one-dimensional family is

$$S_k(x) = B + (M - B) \left[\frac{\sin(k(x - x_0))}{k(x - x_0)} \right]^2, \quad (1.3)$$

where $M > B > 0$ and $k > 0$.

The removable singularity at $x = x_0$ is understood by continuity:

$$\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1.$$

Consequently,

$$S_k(x_0) = M.$$

The function has four useful qualitative properties:

1. a nonzero baseline B ;
2. a unique dominant central maximum at x_0 within the principal lobe;
3. progressively weaker secondary maxima;
4. algebraically decaying oscillatory structure rather than monotone exponential decay.

The secondary maxima of the squared-sinc term occur at nonzero solutions of

$$\tan u = u,$$

with $u = k(x - x_0)$. The first positive solution is approximately

$$u_1 \approx 4.493409,$$

which corresponds to

$$x - x_0 \approx \frac{4.493409}{k} \approx \frac{1.4303\pi}{k}.$$

The next positive solution is approximately $u_2 \approx 7.725252$, giving

$$x - x_0 \approx \frac{7.725252}{k} \approx \frac{2.4595\pi}{k}.$$

These locations are properties of the kernel family and should not be mistaken for experimentally established cognitive distances.

1.8.1 Interpretation of the parameters

The parameters have the following intended meanings:

- B : general competence floor;
- M : peak competence in the core domain;
- k : spatial-frequency parameter controlling how rapidly the profile oscillates in the reduced coordinate;
- x_0 : location of the core specialization;
- D : effective domain width of the modeled skill space;
- E : total modeled resource.

The parameter k is not itself identical to “breadth.” Rather, it controls the width and spacing of the mathematical profile. Interpreting it as a human trait such as transfer width is an empirical mapping that must be tested rather than assumed.

1.9 Exact Finite-Domain Normalization

The strongest mathematical improvement in the present revision is to distinguish the exact finite-domain normalization from the infinite-domain approximation.

Substituting (1.3) into (1.1) gives

$$E = \int_{-D}^D \left[B + (M - B) \left(\frac{\sin(k(x - x_0))}{k(x - x_0)} \right)^2 \right] dx \quad (1.4)$$

$$= 2DB + (M - B)I(k, D, x_0), \quad (1.5)$$

where

$$I(k, D, x_0) = \int_{-D}^D \left(\frac{\sin(k(x - x_0))}{k(x - x_0)} \right)^2 dx. \quad (1.6)$$

Therefore the exact peak height within this model family is

$$\boxed{M = B + \frac{E - 2DB}{I(k, D, x_0)}} \quad (1.7)$$

provided $E > 2DB$.

Equation (1.7), not the infinite-domain expression, is the exact finite-domain result.

1.9.1 Large-domain approximation

If the finite interval contains essentially all of the practically relevant mass of the squared-sinc kernel, then one may use the standard integral

$$\int_{-\infty}^{\infty} \left(\frac{\sin(ku)}{ku} \right)^2 du = \frac{\pi}{k}, \quad k > 0. \quad (1.8)$$

In that regime,

$$I(k, D, x_0) \approx \frac{\pi}{k}, \quad (1.9)$$

and therefore

$$M \approx B + \frac{k(E - 2DB)}{\pi}. \quad (1.10)$$

The original formulation used (1.10) as though it were exact on $[-D, D]$. The correct statement is that it is a large-domain approximation to the exact finite-domain result.

Proposition 1.9.1 (Positivity and feasibility). *Under the conditions $D > 0$, $B > 0$, and $E > 2DB$, the normalized profile defined by (1.7) satisfies $S_k(x) \geq B > 0$ on $[-D, D]$ and has peak $M > B$.*

Proof. The squared-sinc factor is nonnegative. Hence $S_k(x) \geq B$. Since $I(k, D, x_0) > 0$ and $E - 2DB > 0$, equation (1.7) gives $M > B$. $\square$

1.10 What Can and Cannot Be Called “Optimal”

The phrase “optimal cognitive allocation” requires care. A formula that satisfies a resource constraint does not, by itself, prove that it globally optimizes human performance.

What can be established mathematically is a weaker and more defensible statement:

Proposition 1.10.1 (Within-family optimization). *Fix D , E , B , k , and x_0 . Among profiles of the form (1.3), the resource constraint uniquely determines the peak parameter M by (1.7).*

This is a parameter-identification result, not a theorem that the squared-sinc family is globally optimal among all conceivable expertise profiles.

To claim global optimality one would have to specify an objective functional and compare the proposed family with alternative families. For example, one might define

$$\mathcal{J}[S] = \alpha M[S] + \beta \mathcal{T}[S] - \gamma \mathcal{C}[S],$$

where $M[S]$ measures core mastery, $\mathcal{T}[S]$ measures transferable competence, and $\mathcal{C}[S]$ represents maintenance or switching costs. The present paper does not claim to have solved that broader variational problem.

Accordingly, the central claim is now stated as:

Core mathematical claim. The squared-sinc family is a plausible and analytically tractable candidate model for a finite-resource expertise profile with a dominant core, a competence floor, and non-monotonic secondary structure. Its superiority over other kernels is an empirical question, not a theorem established solely by analogy with diffraction.

1.11 The “Side-Lobe” Interpretation

The most distinctive feature of the proposal is the interpretation of the secondary maxima.

A conventional specialization model might predict something like

$$S(x) \downarrow \quad \text{as} \quad d(x, x_0) \uparrow,$$

meaning that competence simply declines with distance from one’s specialty.

The Diffraction Profile instead predicts

$$S(x) \text{ can rise again at selected distances from } x_0.$$

This is the central notion of *structural resonance*.

For example, consider an individual whose core training is in one branch of mathematics. A second field might be institutionally remote yet mathematically resonant because both fields repeatedly employ the same conceptual transformations: invariants, equivalence, operators, symmetry, local-to-global reasoning, dynamical evolution, or obstruction. The model predicts that the person may display competence in the second field that is disproportionately high relative to what ordinary disciplinary distance would suggest.

The claim is not that all distant fields will produce this effect. The claim is that the profile can represent a small number of elevated secondary regions without requiring the person to have independently acquired maximum-depth expertise in each region.

This is the point at which the model goes beyond a T-shape. A T is usually drawn as one vertical and one broad horizontal bar. A diffraction profile permits the horizontal component to be structured: some regions may be much stronger than others because transfer is selective.

Thus a useful conceptual description is

T with side lobes = deep vertical + general competence floor + selected resonant regions.

1.12 Why the Profile Is Not a Literal Physical Wave

The diffraction analogy is structural and mathematical, not ontological.

In physical optics, diffraction emerges from wave propagation and interference. There is no assertion here that thought is a light wave or that neural activity obeys the same boundary-value problem as a slit in an optical screen.

The analogy is useful because the squared-sinc intensity profile has a recognizable and mathematically convenient shape:

dominant maximum + secondary maxima + decaying envelope.

What is being borrowed is the *geometry of the profile*, not the physical mechanism that generates it.

This distinction should remain explicit throughout the paper because otherwise the optical metaphor could be mistaken for a mechanistic neuroscience claim.

1.13 General Kernel Formulation

The squared-sinc function should be viewed as the first member of a broader class of models.

Let $K(r; \theta)$ be a nonnegative kernel of structural distance r with parameter vector θ . Then a

generic expertise profile can be written as

$$S(x) = B + (M - B)K(d(x, x_0); \theta), \quad (1.11)$$

with $K(0; \theta) = 1$ and $0 \leq K \leq 1$ after normalization.

The diffraction hypothesis is the special case

$$K(r; k) = \left(\frac{\sin(kr)}{kr} \right)^2.$$

This generalization is important because it converts the theory from a claim that “human expertise must be a sinc-squared function” into a more reasonable research program:

1. define a structural space;
2. estimate a structural distance;
3. measure competence across that space;
4. compare candidate kernels;
5. determine whether a non-monotone kernel explains the data better than monotone alternatives.

Candidate alternatives could include Gaussian, Laplace, Matérn, mixture, spline, or learned nonparametric kernels. The squared-sinc family is of interest because it produces a mathematically natural sequence of secondary lobes, but it should earn preference through predictive performance rather than through metaphor.

1.14 A Graph and Network Interpretation

Many real intellectual landscapes are not well modeled by one scalar coordinate. Mathematics is especially resistant to a purely linear representation because a single subject can connect strongly to several otherwise distant areas.

Let $G = (V, E_G)$ be a weighted graph whose vertices represent fields, tasks, or conceptual modules. Let $w_{ij} \geq 0$ denote the strength of the structural connection between vertices i and j .

A graph-based expertise profile can be written schematically as

$$S_i = B + (M - B)K_G(i, i_0), \quad (1.12)$$

where K_G is a graph kernel centered at the core node i_0 .

One option is a heat kernel, which would generally produce monotone decay in graph distance. A different option could incorporate spectral oscillation and produce secondary maxima. The precise choice is an open empirical question.

This observation is useful because the companion mathematical-landscape preprint already describes mathematical specialization as a network with high-transfer interfaces. The present paper therefore supplies a theoretical reason why a specialist may develop particularly strong “side lobes” at a few nonadjacent but structurally connected nodes.

1.15 Connection to a T-Shaped Career with Side Lobes

The model was motivated in part by a practical observation: some intellectual careers do not resemble a clean T. They have one dominant vertical, but also a handful of unexpectedly strong secondary regions.

A person may, for example, have:

- one domain of unusually deep expertise;
- broad literacy across many adjacent areas;
- two or three secondary domains in which competence is substantially higher than a simple distance-decay model would predict.

The ordinary T metaphor would show only the first two properties. The Diffraction Profile attempts to represent all three.

The distinction between *independent specialization* and *transferred competence* is crucial. A secondary lobe is not equivalent to a second vertical. The model is instead intended to represent a domain that becomes unusually accessible because the individual’s core training has built reusable representations.

This gives a formal interpretation to a phrase such as “T-shaped with side lobes.” It is not merely a more elaborate career metaphor. It is a hypothesis that the distribution of competence across domains may contain structured secondary maxima.

1.16 Research Predictions

The revised model produces several predictions that are stronger than broad statements such as “deep experts are versatile.”

1.16.1 Prediction 1: Non-monotonic transfer

If structural distance can be operationalized, competence should not always decay monotonically with that distance. At least some experts should show measurable secondary peaks.

1.16.2 Prediction 2: Resonant rather than random secondary peaks

Secondary peaks should be more likely in domains that share latent structure with the core than in equally distant domains selected at random.

1.16.3 Prediction 3: Core depth and side-lobe structure are partially separable

A person can have very high core mastery M without having all secondary lobes equally high. Therefore peak depth and transfer pattern should be treated as different variables.

1.16.4 Prediction 4: Individual profiles should differ

Two people with comparable core mastery can have different side-lobe locations because their previous learning histories, representations, and experiences differ.

1.16.5 Prediction 5: Structural similarity should outperform superficial similarity

When predicting transfer, measures based on shared underlying operations or representations should outperform measures based solely on topic labels or vocabulary overlap.

1.16.6 Prediction 6: Deepening a core can move the profile

As expertise develops, the side-lobe structure may change. Transfer is therefore predicted to be dynamic rather than fixed over the lifetime.

1.16.7 Prediction 7: T-with-side-lobes should be empirically distinguishable from a conventional T

A conventional T can be represented approximately by one dominant region plus a broad low plateau. The diffraction hypothesis predicts localized secondary elevations above that plateau. Those elevations should be detectable with sufficiently rich measurement.

1.17 The Polymath Resonance Effect

The original paper called the first prediction the *Polymath Resonance Effect*. The label is retained, but its meaning is narrowed.

Definition 1.17.1 (Polymath Resonance Effect). The Polymath Resonance Effect is the proposed tendency for an individual with unusually deep expertise in one domain to exhibit disproportionately high competence in selected other domains that possess strong structural correspondence with the core domain, even when the two domains are not ordinarily classified as adjacent.

The claim is not that famous polymaths necessarily follow a squared-sinc curve. Historical polymathy is too heterogeneous for such a statement. The stronger claim—and the one that can eventually be tested—is that some experts should show measurable, structured secondary competence that is not adequately explained by uniform broad ability or by simple topical proximity.

1.18 Educational and Career Implications

If the model is approximately correct, several practical consequences follow.

1.18.1 Do not maximize the number of deep peaks by default

If every deep specialization requires substantial maintenance, then attempting to create many independent peaks may reduce the time available to deepen the core and may also reduce the opportunity for transfer to emerge naturally.

This does not mathematically prove that one peak is always superior. It implies that multiple peaks should be evaluated against their maintenance cost and their degree of structural coupling.

1.18.2 Choose a core that generates representations with broad transfer

A core domain is strategically valuable not only because it offers a research identity, but because it can reorganize how the learner represents later problems.

The companion mathematical-landscape paper makes this argument explicitly for a topology/homotopy/homological trajectory: repeated exposure to objects, maps, equivalences, invariants, obstructions, functoriality, and local-to-global reasoning is hypothesized to alter the learner's future mathematical representation system.

1.18.3 Treat secondary domains as resonant investments

A side lobe should not be interpreted as “learn a little of everything.” It is the opposite. The theory predicts that breadth is most valuable when it is selective and structurally connected.

1.18.4 Reserve maintenance effort strategically

The model supports a simple budget intuition. If c_i denotes the ongoing maintenance cost of a secondary domain and B_T denotes the total maintenance budget, then

$$\sum_i c_i \leq B_T.$$

The objective is therefore not to accumulate the largest possible number of subject labels. It is to select regions whose maintenance produces disproportionate transfer and optionality.

1.19 Implications for Human–AI Collaboration

The model becomes especially relevant when technical mathematical labor becomes increasingly automatable.

The strategic question is not whether a given mathematical field is permanently “AI-proof.” That target is unstable. Instead, the relevant question is whether deep study changes the human’s representational capacity in ways that remain valuable when machines become more effective at computation, proof search, formalization, retrieval, and verification.

Within the companion mathematical-landscape framework, the proposed division of labor is approximately

human: problem selection, structural intuition, interpretation, synthesis, judgment

and

machine: search, computation, formalization, routine verification, exploration.

The Diffraction Profile offers a conceptual reason for this division. If deep expertise creates transferable representations, then AI can be used not merely to replace technical effort but to expand the number of problems that those representations can be applied to.

A robust human–AI loop is therefore

Human–AI research loop

human question $\rightarrow$ AI exploration
 $\rightarrow$ formal/computational checking
 $\rightarrow$ human interpretation $\rightarrow$ revised question.

This is consistent with the companion landscape paper’s central recommendation: deep human

conceptual training combined with aggressive AI augmentation rather than human competition with machines at technical execution.

1.20 Operationalizing Structural Distance

The hardest empirical problem is turning the phrase “structural distance” into a measurable variable.

Several strategies are possible.

1.20.1 Psychometric task embeddings

Give participants a battery of tasks spanning multiple domains. Represent each task by observable features such as operations required, abstraction level, transformation type, memory burden, symbolic structure, and inference pattern. Estimate a distance matrix from those features.

1.20.2 Knowledge-graph distance

Represent domains as nodes in a graph whose edges indicate shared definitions, methods, invariants, proof patterns, or prerequisite structures. Structural distance could be defined using weighted shortest paths or spectral distances.

1.20.3 Semantic embeddings

Large language models or domain-specific embedding systems could be used to produce candidate representations of mathematical or professional concepts. These representations should be treated as proxies, not as ground truth. Their usefulness would have to be compared against human judgments and task-level behavioral data.

1.20.4 Human-judged structural similarity

Experts could directly rate how much two tasks share their underlying solution architecture. Such judgments provide an independent test against purely machine-generated similarity metrics.

1.20.5 O*NET-style occupational embeddings

For professional skills, standardized occupation and task taxonomies can provide another approximation to structural distance. The original paper suggested O*NET cosine similarity as one possible route. The present formulation retains this as an exploratory method rather than as a validated measurement instrument.

1.21 Operationalizing Skill Depth

A second difficult problem is the meaning of $S(x)$.

Skill depth can be measured in many non-equivalent ways:

- accuracy on representative tasks;
- time to solution;
- ability to solve novel problems;
- ability to explain and generalize solutions;
- transfer to unfamiliar tasks sharing the same structure;
- formal proof competence;
- research-level problem formulation.

A serious empirical study should therefore define a task-specific competence function rather than treating a resume label or years of experience as a direct measure of $S(x)$.

The same caution applies to M and B . They are model parameters whose empirical interpretation must be fixed before fitting.

1.22 Empirical Falsification Strategy

The theory should be considered successful only if it survives comparison with simpler alternatives.

A possible experimental workflow is:

1. Select a group of experts with identifiable core specializations.
2. Measure performance across a broad set of tasks.
3. Estimate structural distances among tasks using independent methods.
4. Fit a monotone decay model.
5. Fit the diffraction kernel model.
6. Fit flexible nonparametric or mixture models.
7. Compare predictive accuracy on held-out tasks.
8. Test whether predicted secondary maxima replicate across cohorts.

The crucial comparison is not “does the graph look like a diffraction pattern?” It is whether the diffraction family predicts held-out competence better than simpler hypotheses.

For example, one could compare

$$S_{\text{mono}}(r) = B + (M - B)e^{-\lambda r}$$

against

$$S_{\text{diff}}(r) = B + (M - B) \left(\frac{\sin(kr)}{kr} \right)^2$$

and against a flexible spline or Gaussian-process model.

A secondary-lobe hypothesis is supported only if it improves predictive performance in a manner that is replicable and not merely produced by overfitting.

1.23 Statistical Considerations

The model contains several parameters, and overparameterization is a concern. In particular, a flexible embedding combined with a flexible kernel can produce apparent secondary structure even when none exists.

Therefore an empirical study should use:

- preregistered structural features where feasible;
- out-of-sample prediction;
- cross-validation or held-out cohorts;
- explicit comparison with monotone baselines;
- uncertainty intervals for fitted parameters;
- sensitivity analysis under alternative definitions of structural distance.

A useful practical criterion is whether the location of secondary competence remains stable when reasonable alternative embeddings of the task space are used.

1.24 Illustrative Interpretation of the Parameter k

The original paper described k as a “transfer-width coefficient.” That phrase is potentially misleading because a larger k in the squared-sinc family makes the central lobe narrower and places secondary maxima closer together in the one-dimensional coordinate.

Accordingly, the present paper calls k the *profile-frequency parameter*. Its cognitive interpretation is left open until the coordinate system has been empirically defined.

One may later define a derived breadth measure, for example the half-width at half-maximum $w_{1/2}$ of the central lobe, or another quantile of the kernel mass. Such derived measures are often more interpretable than k itself.

This distinction prevents the model from conflating a mathematical parameter with a psychological construct.

1.25 Alternative Parameterization by Core Width

For some empirical applications it may be preferable to parameterize the model by a core-width scale σ rather than by k . Since the central lobe width is proportional to $1/k$, one may write

$$\sigma \propto \frac{1}{k}.$$

Then a narrower core corresponds to smaller σ and, within the same normalized family, greater peak concentration.

The exact proportionality depends on which width definition is chosen. For example, the first zero of the sinc-squared factor occurs at

$$k|x - x_0| = \pi,$$

so the half-width of the main zero-to-zero lobe is π/k on either side of the core.

This provides an empirically interpretable scale without pretending that k itself is a psychological variable.

1.26 Multiple Cores and Genuine Polymathy

The original paper proposed multi-slit diffraction as a possible extension for rare multi-domain polymaths. That idea is retained, but formalized more cautiously.

A simple superposition is

$$S(x) = B + \sum_{j=1}^m A_j K(d(x, x_j); \theta_j), \quad (1.13)$$

subject to a global resource constraint such as

$$\int S(x) dx = E.$$

In the multi-core case, however, interference between the components depends on the chosen representation, and there is no general theorem that a simple sum is the psychologically correct model. The equation should therefore be treated as a candidate extension, not as a derived

result.

The interesting empirical question is whether apparent multiple peaks are better explained by genuinely independent cores or by a single core with resonant side lobes.

1.27 Dynamic Expertise Profiles

Expertise changes over time. Let

$$S(x, t)$$

denote competence at time t .

The static model can then be extended conceptually by allowing

$$B = B(t), \quad M = M(t), \quad k = k(t), \quad x_0 = x_0(t).$$

A simple dynamic resource equation could be written as

$$\frac{\partial S}{\partial t} = G(x, t) - L(x, t), \tag{1.14}$$

where G represents learning and L represents forgetting, maintenance cost, or opportunity cost.

This framework could eventually distinguish three processes:

$$\text{acquisition}, \quad \text{transfer}, \quad \text{maintenance}.$$

The original static model treats them implicitly. A dynamic theory would treat them separately.

1.28 Why Secondary Lobes May Matter for Research Strategy

The practical significance of a side lobe is not that it creates a second field of equal prestige or equal depth. Its value is that it can create a high-transfer interface.

Suppose a researcher has a central expertise in domain A and a resonant side lobe in domain B . The research opportunity may lie not in becoming independently maximal in B , but in exploiting the interface

$$A \leftrightarrow B.$$

This is precisely why the companion mathematical-landscape preprint distinguishes a *cognitive home* from a *research home*. A strong vertical can generate representations that make a particular interface unusually productive without requiring the researcher to treat every interface as a separate career.

In mathematics, for example, a topology/homotopy core may create unusually valuable interfaces to dynamics, geometry, category theory, and homological structures. The statistics preprint then provides a quantitative sense of the size of the communities occupying those regions.

1.29 Failure Modes

1.29.1 Failure mode 1: The diffraction metaphor becomes literal

The optical analogy can be mistaken for a neural mechanism. This is a category error. The model borrows a functional form, not a physical substrate.

1.29.2 Failure mode 2: The profile is treated as universally optimal

A normalized function is not an optimization theorem. The present paper therefore restricts its optimization language to within-family normalization and parameter identification.

1.29.3 Failure mode 3: Structural distance is defined after seeing the results

This would make the hypothesis unfalsifiable. Structural distance must be specified independently enough that the model can make predictions.

1.29.4 Failure mode 4: Any broad competence is called a side lobe

A side lobe must be a localized, reproducible elevation relative to an explicit baseline and an explicit structural-distance model.

1.29.5 Failure mode 5: Topic similarity is confused with structural similarity

Two domains can share vocabulary while requiring very different reasoning architectures, and two domains with different vocabulary can share deep structure. The theory concerns the latter.

1.29.6 Failure mode 6: A personal profile is generalized too quickly

The theory may have originated in introspection or lived experience. That is a legitimate source of hypotheses but not sufficient evidence for generality. The model becomes scientific only when it generates predictions about other individuals.

1.29.7 Failure mode 7: Side lobes become an excuse for excessive breadth

The model does not prescribe maximal breadth. It predicts selective transfer under resource constraints. Excessive breadth with insufficient depth remains a failure mode.

1.30 Testable Research Program

The following sequence would turn the current theoretical paper into an empirical program.

1.30.1 Study A: Mapping expertise

Measure performance across a sufficiently rich task set for individuals with a clearly identified core specialization.

1.30.2 Study B: Estimating structural distance

Construct at least two independent distance matrices, for example human-expert judgments and a task-embedding method.

1.30.3 Study C: Kernel comparison

Fit monotone and non-monotone kernels, including the proposed squared-sinc family.

1.30.4 Study D: Out-of-sample validation

Evaluate whether the predicted side-lobe locations improve held-out prediction.

1.30.5 Study E: Longitudinal dynamics

Track people over time to determine whether deepening the core changes the location or amplitude of secondary competence.

1.30.6 Study F: Career outcomes

Test whether a T-with-side-lobes configuration predicts research optionality, interdisciplinary output, or human–AI leverage better than simpler measures of breadth and depth.

1.31 Relation to Education and Human Capital

The model has implications for educational design, but those implications should be stated as contingent recommendations rather than prescriptions.

If a domain creates highly reusable representations, deep study may have benefits outside the domain itself. If the learner then selects a small number of resonant neighboring regions, those investments may compound. This suggests a curriculum strategy of

deep core + selective transfer + broad literacy

rather than

maximal number of shallow subjects.

The companion mathematics paper instantiates this idea as a particular architecture. It should not be read as a proof that its proposed vertical is universally optimal. Its role is to provide a concrete case in which the theory can be made operational.

1.32 What the Model Means for a Person with a “T with Side Lobes” Profile

A useful interpretation of the framework for an individual is not “I am good at many things.” It is more specific:

I have one domain in which my representations have been made unusually deep, a broad competence floor across many domains, and a small set of secondary domains in which that deep representation appears to transfer unusually well.

The purpose of the mathematical model is to convert this qualitative description into a structure that can eventually be measured.

The theory therefore gives language to an intellectual development pattern that is neither pure specialization nor indiscriminate polymathy. It is specialization whose effects radiate selectively.

1.33 Discussion

The central conceptual contribution of the Diffraction Profile is the claim that the geometry of expertise may be more informative than categorical labels such as “specialist”, “generalist”, or “T-shaped professional.” A continuous profile can express several facts simultaneously: depth, breadth, selective transfer, and non-monotonic competence.

The mathematical form should not be oversold. A squared-sinc curve is a candidate model. Its value depends on whether it predicts data. The strongest version of the theory is therefore not the statement that “human expertise is diffraction.” The strongest version is the following research hypothesis:

Research hypothesis. Expertise may be distributed over a structured domain space in such a way that a dominant core specialization produces selective secondary concentrations of competence at structurally resonant locations. A continuous, non-monotone kernel with a dominant central peak and secondary lobes may provide a more faithful description of such profiles than discrete I-, T-, Pi-, or Comb-shaped typologies alone.

This hypothesis is useful because it is vulnerable to evidence. A sufficiently rich dataset could show that transfer is monotone, that secondary peaks are random, that simple Gaussian kernels outperform the diffraction kernel, or that structural distance contributes little beyond general ability and prior experience. Any of those outcomes would force revision.

Conversely, repeated evidence of stable, structure-predictable secondary competence would support the main idea and justify more elaborate models.

1.34 Limitations

The present theory remains speculative in several important respects.

First, $S(x)$ is not yet a validated psychometric quantity. Second, the structural space itself remains to be operationalized. Third, the finite resource budget E is a modeling abstraction. Fourth, the squared-sinc family is selected for mathematical tractability and conceptual resemblance to diffraction, not because a comparative model-selection study has established it as uniquely superior. Fifth, the current one-dimensional profile may be inadequate for genuinely networked skill spaces. Sixth, the model does not account explicitly for motivation, affect, social environment, institutional constraints, health, or economic resources. Seventh, the theory does not distinguish all possible forms of transfer, including transfer that is negative, context-dependent, or mediated by explicit instruction.

The model also does not yet establish that a deep core is always preferable to multiple cores. That claim depends on the objective function, the cost of maintenance, and the structure of transfer. A genuine multi-core professional may be optimally described by a different model.

Finally, there is a risk of post hoc storytelling: once a person is known to have strengths in several fields, it is easy to construct a structural story explaining them. The proposed empirical program must therefore be predictive and preferably preregistered.

1.35 Conclusion

The Diffraction Profile provides a mathematical language for a particular conception of expertise: one deep center, a nonzero generalist floor, and selected secondary regions of enhanced competence produced by structural resonance. The central contribution is not the claim that optics literally describes cognition. It is the proposal that a continuous, resource-constrained, non-monotone profile may capture a class of intellectual careers that categorical I-, T-, Pi-, and Comb-shaped descriptions leave underspecified.

The revised formulation makes four commitments explicit. First, the resource constraint is a model of finite investment, not an established biological conservation law. Second, the finite-domain normalization is exact and the familiar π/k expression is treated as a large-domain

approximation. Third, the squared-sinc family is not claimed to be globally optimal; only its within-family normalization is derived exactly. Fourth, the one-dimensional profile is a reduced model of a more general structured skill space.

With those qualifications, the theory yields a coherent research program. It predicts non-monotonic transfer, structurally localized secondary competence, individual differences in side-lobe structure, and a potential distinction between independent specialization and transferred competence. It also supplies a theoretical foundation for the companion mathematics preprint, in which a particular T-shaped architecture is proposed, and for the companion statistical preprint, which maps the approximate size of the research communities occupying that architecture.

The resulting research program can therefore be summarized as

$$\boxed{\text{Theory} \longrightarrow \text{Architecture} \longrightarrow \text{Ecosystem}}$$

with the present paper providing the first layer.

The strongest future test is empirical: construct a meaningful structural space, measure expertise across it, compare monotone and non-monotone models, and determine whether secondary competence really appears where the theory predicts. If it does, the Diffraction Profile becomes more than a metaphor. If it does not, the model should be replaced or substantially modified.

Appendix: Mathematical and Computational Details

1.36 Mathematical Details of the Squared-Sinc Kernel

For completeness, define

$$f(u) = \left(\frac{\sin u}{u} \right)^2, \quad f(0) = 1.$$

Then

$$S(x) = B + (M - B)f(k(x - x_0)).$$

The zeros occur at

$$k(x - x_0) = n\pi, \quad n \in \mathbb{Z} \setminus \{0\}.$$

The extrema away from the origin satisfy

$$f'(u) = 0 \iff u \cos u - \sin u = 0,$$

which is equivalent to

$$\tan u = u.$$

Hence the secondary maxima are determined by the positive roots of $\tan u = u$.

The total mass over the real line satisfies

$$\int_{-\infty}^{\infty} f(ku) du = \frac{1}{k} \int_{-\infty}^{\infty} f(v) dv = \frac{\pi}{k}.$$

This identity underlies the large-domain approximation used in the main text.

1.37 A Useful Comparison: Monotone versus Resonant Transfer

A monotone comparison model is

$$S_{\text{mono}}(x) = B + (M - B) \exp\{-\lambda d(x, x_0)\}, \quad (1.15)$$

where $\lambda > 0$.

The monotone model predicts a continuous decline in competence as structural distance increases. The diffraction model predicts zeros and secondary maxima in the kernel component. Therefore the two models make qualitatively different predictions even when both have the same core peak and baseline.

This distinction is experimentally useful because the theory is most interesting precisely when it can be distinguished from a simpler monotone decay law.

1.38 Illustrative Figure with Fixed Peak

Figure 1.1 compares two normalized shapes with the same visible peak and baseline. The purpose is purely geometric: to show that changing the frequency parameter changes the location and spacing of secondary structure. It is not an optimization demonstration.

1.39 Potential Extension to Learned Kernels

An empirical version of the theory could estimate a kernel directly from data. Suppose observed performance is Y_i for participant p on task i , and each task has a structural coordinate z_i . One could model

$$Y_i = \alpha_p + \beta_p K(z_i, z_{0p}) + \varepsilon_i,$$

with participant-specific core z_{0p} and residual ε_i .

The key model-selection question would be whether K is better represented by a monotone kernel, a multimodal kernel, the diffraction family, or a flexible learned function. This would turn the present theoretical argument into a conventional statistical model-comparison problem.

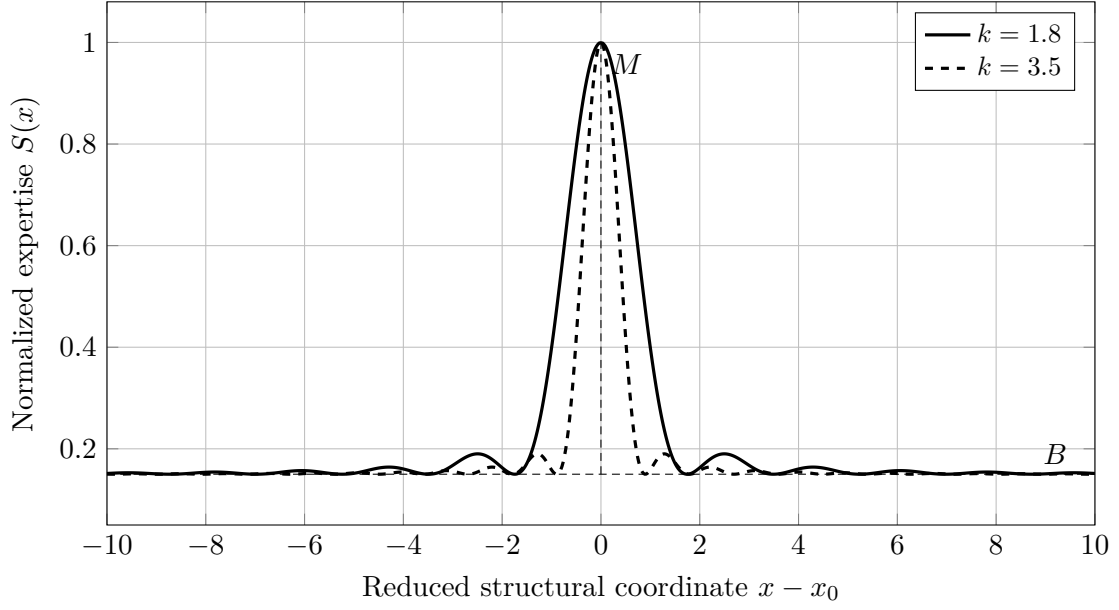


Figure 1.1: Illustrative squared-sinc profiles with a common baseline and normalized peak. The figure shows shape changes as k changes; it is not evidence that a larger k is psychologically superior.

1.40 Potential Extension to Research Networks

For the mathematical application, let vertices be mathematical subfields and let weighted edges represent conceptual transfer. A core field i_0 could generate a predicted competence vector

$$\mathbf{S} = B\mathbf{1} + (M - B)\mathbf{K}_{i_0},$$

where $\mathbf{K}_{i_0}$ is a graph kernel centered at i_0 .

A side lobe then corresponds to a field j for which

$$K_{i_0j}$$

is substantially larger than would be predicted from simple graph distance alone. This formulation provides a natural bridge from the one-dimensional theoretical model to the mathematical architecture in the companion preprint.

1.41 Terminological Clarifications

To keep later versions precise, the following terminology is recommended:

- use *resource constraint* rather than literal biological energy;
- use *profile-frequency parameter* for k rather than assuming k directly equals transfer width;

- use *structural distance* only after specifying how it is measured;
- use *secondary competence* or *side lobe* for predicted localized competence elevations;
- use *candidate optimum* or *within-family optimum* unless a global objective has been explicitly solved;
- reserve *intelligence* for the broad conceptual framing, not as a claim of a complete psychometric theory.

1.42 Author's Note

This manuscript is intentionally written as a theoretical preprint rather than as a declaration that the model has already been empirically validated. Its purpose is to make a personal and strategic intuition mathematically explicit enough that other researchers can attack it, test it, improve it, or reject it. The strongest version of the idea is therefore not that every intelligent person must have a diffraction profile. It is that a diffraction-like profile is a useful candidate hypothesis for a class of expertise distributions in which deep specialization generates selective, structurally organized transfer.

2. Research-Ecosystem Size in Mathematics

2.1 Research-Ecosystem Statistics: Scope and Purpose

Research Ecosystem Size in Mathematics

A quantitative guide to the approximate size of mathematical research communities

Pearl Bipin Pulickal

September 2026

Purpose.

This document has one purpose: to provide a quantitative sense of how large different mathematical research communities are.

The objective is not to claim that there is an exact global census of algebraic topologists, number theorists, ergodic theorists, higher category theorists, or other narrow specialties. No sufficiently standardized global census of that kind was identified.

Instead, the report combines broad bibliometric evidence with explicitly labeled field-level estimates so that a reader can distinguish approximately between communities of hundreds, thousands, and tens of thousands of researchers.

2.2 What Does “How Many Researchers?” Mean?

The phrase “number of researchers in a field” is ambiguous.

At least four different populations can be considered:

$$N_{\text{indexed}} = \text{all authors ever indexed};$$
$$N_{\text{active}} = \text{authors publishing in a specified calendar year};$$
$$N_{>2} = \text{authors active in that year with more than two publications overall};$$

and

$N_{>9}$ = authors active in that year with more than nine publications overall.

These are very different populations.

Hulek and Teschke’s analysis of author-disambiguated ZBMATH data shows that about 44% of indexed authors are associated with only one publication. The median author has two publications and the mean is approximately 7.9 publications [10].

Thus:

$$N_{\text{indexed}} \gg N_{\text{established}}.$$

A simple bibliographic author count should therefore not be interpreted as the number of people currently pursuing a mathematical research career.

2.3 The Main Large-Scale Statistical Source

The principal broad quantitative source used in this report is the author-oriented analysis of ZBMATH by Klaus Hulek and Olaf Teschke [10, 11].

The analysis uses author-disambiguated ZBMATH data and the Mathematics Subject Classification (MSC). The authors report that only about 3.5% of authorships remained ambiguous in the dataset used for their analysis, meaning that approximately 96.5% of authorships were sufficiently disambiguated for author-level analysis [11].

The same work measures actively publishing authors by calendar year and groups mathematics into ten broad MSC-based clusters.

The important methodological point is:

field membership is overlapping rather than disjoint.

A researcher can simultaneously publish in algebra, number theory, algebraic geometry, topology, dynamics, probability, category theory, and other adjacent areas.

Therefore the numbers below should be interpreted as *research-community size indicators*, not as a partition of all mathematicians into mutually exclusive boxes.

2.4 Broad Mathematical Scale

The ZBMATH author statistics show that the number of calendar-year active mathematical authors reached the order of 10^5 by the end of the historical series. The published graph places the 2020 endpoint at roughly the 1.8×10^5 scale [10].

Thus a useful global order-of-magnitude statement is

$$N_{\text{active mathematics}} \approx 10^5 - 2 \times 10^5.$$

This is a database-defined population of calendar-year-active mathematical authors. It is *not* the number of professional mathematicians in the world.

Important distinction.

The 10^5 -scale figure should be read as the size of a large bibliometric publication ecosystem, not as a census of everyone whose occupation is “mathematician.”

Conversely, the actual global number of people doing mathematical research may differ because some researchers publish outside the database or have different publication profiles.

2.5 Broad MSC Research Clusters

Hulek and Teschke use ten broad clusters. The principal clusters relevant to this report are:

Cluster	Principal content
NTAG	Number Theory + Algebra + Algebraic Geometry + Group Theory
TopGeom	Topology + Geometry
Ana	Real + Complex Analysis
OpTh	Harmonic + Functional Analysis + Operator Theory
DIEq	Differential/Integral Equations + Dynamical Systems
ProbStat	Probability + Statistics
Disc	Discrete Mathematics + Convex Geometry
OptCS	Optimization + Numerical Analysis + Computer Science
MaPh	Mathematical Physics
Gen	General Mathematics + Foundations

The published historical figures show that several of these broad clusters reach the *tens-of-thousands* scale by the end of the series [11].

This establishes a useful hierarchy:

major mathematical ecosystems can be five-figure communities.

The narrower question is how those broad communities decompose into specialist fields.

2.6 Quantitative Orientation Table

The following table is the main statistical summary.

The ranges are deliberately written as order-of-magnitude bands rather than apparently exact point estimates.

Field	Approximate global ecosystem size	Grade	Interpretation
Number Theory	High thousands to low tens of thousands	C	Large global ecosystem. The direct ZB-MATH cluster is broader because NTAG also contains algebra, algebraic geometry, and group theory.
Abstract Algebra	Many thousands	B/C	Large ecosystem spread across several MSC algebra classes. No single disjoint “abstract algebra” headcount exists.
Group Theory	Several thousand	C	Substantial international community within NTAG, with strong overlap with geometry, topology, dynamics, and algebra.
Representation Theory	Several thousand	C	Established international community overlapping algebra, Lie theory, geometry, number theory, and category theory.
Lie Theory	Low thousands	C	Specialist but substantial ecosystem distributed across Lie groups, Lie algebras, representation theory, and geometry.
Algebraic Geometry	Several thousand to high thousands	B/C	Major international ecosystem. Exact isolated size is obscured by overlap with number theory, topology, representation theory, and algebra.
Algebraic Topology	Few thousand	C	Specialist component of the much larger TopGeom ecosystem. Exact isolated count is not given by the broad ZBMATH cluster analysis.
Homotopy Theory	Low thousands	C/D	Narrower component of algebraic topology with substantial cross-listing into homological algebra, category theory, geometry, and higher structures.
Homological Algebra	Low thousands	C/D	Crosses algebra, category theory, algebraic topology, and algebraic geometry. Exact disjoint population is not directly measurable from broad MSC clusters.
Category Theory	Low thousands	C	Substantial specialist ecosystem, with strong overlap with homological algebra, algebra, topology, logic, and computer science.
Higher Category Theory	Hundreds to low thousands	D	Specialist ecosystem. No authoritative global census identified. This is an ecosystem-scale inference, not a measured population.

Field	Approximate global ecosystem size	Grade	Interpretation
Higher Algebra	Hundreds to low thousands	D	Specialist ecosystem strongly overlapping higher category theory, homotopy theory, derived geometry, and representation theory.
Homotopy Type Theory	Hundreds; potentially approaching low thousands under broad inclusion	D	Small specialist community. No defensible global active-researcher census identified.
Dynamical Systems	Several thousand	C	MSC 37 belongs to the larger DIEq cluster. Dynamics is therefore a major specialist ecosystem nested within a still larger differential-equations community.
Ergodic Theory	Low thousands	C/D	Strong specialist ecosystem embedded across dynamics, measure theory, probability, harmonic analysis, and smooth dynamics.
Complex Dynamics	Hundreds to low thousands	C/D	Intersection of complex analysis and dynamics; field boundaries are highly porous.
Arithmetic Dynamics	Hundreds to low thousands	D	Highly cross-listed among number theory, algebraic geometry, arithmetic geometry, and dynamics. No authoritative census identified.
Harmonic Analysis	Several thousand	B/C	Substantial mathematical community within the broader OpTh cluster.
Functional Analysis	Several thousand	B/C	Substantial component of the OpTh cluster, overlapping heavily with operator theory and harmonic analysis.
Measure Theory	Several thousand	C	Distributed over measure/integration, analysis, probability, and ergodic theory.
Measure-Theoretic Probability	Several thousand	C	Mathematical core of the much larger probability/statistics ecosystem, with strong overlap with stochastic processes and ergodic theory.
Mathematical Logic	Several thousand	C	Large enough to support multiple international specialist communities, including model theory, set theory, computability, proof theory, and type theory.

2.7 Evidence Grades

Grade Type		Meaning
A	Direct count	Reproducible database query under an explicit field and activity definition.
B	Broad database evidence	A database gives a strong measured population, but the category is broader than the field being discussed.
C	Triangulated estimate	The value is inferred from broad bibliometric clusters, MSC structure, publication patterns, and specialist-community evidence.
D	Community proxy	No defensible global census was found. The band communicates only the approximate scale of the specialist ecosystem.

How to read C and D.

“Higher Category Theory: hundreds to low thousands” does not mean that a database query directly returned that interval.

It means that the available evidence supports a specialist community of roughly that order of magnitude, while the exact boundary of the field is too unstable for a defensible census.

2.8 What the Numbers Actually Tell Us

The purpose of the table is not to produce a league table.

It answers the practical question:

Roughly how many people are working in this mathematical ecosystem?

A useful scale interpretation is:

Approximate scale	Interpretation
10^5 scale	Very large mathematical ecosystem; major global research civilization.
10^4 scale	Large international ecosystem with extensive subdivision and many research niches.
10^3 scale	Substantial specialist field; enough international activity for many research directions.
10^2 scale	Small specialist community; usually best understood as a niche attached to a larger umbrella.
Below 10^2	Very small specialist niche; exact demographic statistics become particularly unstable.

Thus the difference between

$$300, \quad 3,000, \quad 30,000$$

is strategically meaningful even when an exact census is impossible.

2.9 Broad Ecosystem versus Specialist Niche

The numerical pattern is approximately:

$$\boxed{\text{broad mathematical civilization} \sim 10^4\text{--}10^5}$$

$$\boxed{\text{strong specialist field} \sim 10^3\text{--several} \times 10^3}$$

$$\boxed{\text{narrow specialist niche} \sim 10^2\text{--}10^3.}$$

For example:

$$\text{Topology} + \text{Geometry}$$

contains broad communities which then subdivide into:

$$\text{Topology}, \quad \text{Differential Geometry}, \quad \text{Algebraic Geometry}, \quad \text{Symplectic Geometry},$$

and narrower structures such as

$$\text{Algebraic Topology} \rightarrow \text{Homotopy Theory} \rightarrow \text{Stable Homotopy} \rightarrow \text{Chromatic Homotopy}.$$

Likewise:

$$\text{Dynamical Systems} \rightarrow \text{Ergodic Theory}$$

and:

Algebra $\rightarrow$ Representation Theory $\rightarrow$ Geometric Representation Theory.

Population generally becomes smaller as specialization becomes narrower.

2.10 Why the Numbers Cannot Simply Be Added

Suppose one writes

$$N_{\text{Number Theory}} + N_{\text{Algebra}} + N_{\text{Algebraic Geometry}}.$$

This does *not* give the number of unique researchers in the union.

A single researcher may belong to all three.

Hulek and Teschke explicitly note that their area-specific active-author counts are calculated separately and that researchers active in multiple clusters can therefore appear more than once in the cumulative display. They estimate that this overlap raises the cumulative height by about 20% relative to the corresponding non-overlapping author count [11].

Therefore:

$$N(A \cup B) = N(A) + N(B) - N(A \cap B).$$

The intersection terms are not available directly from the broad published cluster figures.

Thus the field estimates in this report should *not* be summed to produce a supposed global researcher count.

2.11 Publication Activity Is Highly Skewed

Hulek and Teschke report approximately

$$44\%$$

of indexed authors with only one publication, while

$$\text{median}(P) = 2$$

and

$$\text{mean}(P) \approx 7.9.$$

This means:

all-time indexed authors

can greatly exceed

currently established research population.

Their use of the > 2 and > 9 publication thresholds is therefore statistically useful for revealing a more stable research core [10].

A useful conceptual hierarchy is:

N_1 = broad publication participation, N_3 = more stable research participation, N_{10} = established publishing core.
--

These thresholds are reference points, not universal definitions.

2.12 Calendar-Year Activity versus Career Membership

One can distinguish

$$N_{\text{career}}$$

from

$$N_{\text{calendar active}}.$$

Someone may have published several mathematical papers years ago and now work primarily in another area.

Conversely, a young researcher may have only a few publications while being an active member of a research community.

Therefore no single statistic perfectly captures “number of researchers.”

For this report, “ecosystem size” means approximately

people visibly participating in the research literature of the field.

It does not claim to count all students, all postdocs, unpublished researchers, retired researchers, or industry researchers.

2.13 Field-by-Field Statistical Summary

Field	Approximate scale	Community interpretation	Grade
Number Theory	10^4 -scale	Large global field	C
Abstract Algebra	10^4 -scale	Large broad ecosystem	B/C
Group Theory	Few $\times 10^3$	Large specialist field	C
Representation Theory	Few $\times 10^3$	Large specialist field	C
Lie Theory	10^3 -scale	Specialist but established	C
Algebraic Geometry	10^3 – 10^4	Major global field	B/C
Algebraic Topology	Few $\times 10^3$	Established specialist field	C
Homotopy Theory	10^3 -scale	Specialist field	C/D
Homological Algebra	10^3 -scale	Specialist bridge field	C/D
Category Theory	10^3 -scale	Established specialist ecosystem	C
Higher Category Theory	10^2 – 10^3	Small specialist ecosystem	D
Higher Algebra	10^2 – 10^3	Small specialist ecosystem	D
Homotopy Type Theory	10^2 , potentially approaching 10^3	Small specialist ecosystem	D
Dynamical Systems	Few $\times 10^3$	Large specialist field	C
Ergodic Theory	10^3 -scale	Specialist field with broad interfaces	C/D
Complex Dynamics	10^2 – 10^3	Specialist intersection field	C/D
Arithmetic Dynamics	10^2 – 10^3	Specialist intersection field	D
Harmonic Analysis	Few $\times 10^3$	Established specialist field	B/C
Functional Analysis	Few $\times 10^3$	Established specialist field	B/C
Measure Theory	Few $\times 10^3$	Distributed analytical ecosystem	C
Measure-Theoretic Probability	Few $\times 10^3$	Strong mathematical probability ecosystem	C

Field	Approximate scale	Community interpretation	Grade
Mathematical Logic	Few $\times 10^3$	Established broad specialist ecosystem	C

2.14 Research Community Density

A useful empirical heuristic is:

population decreases as specialization becomes narrower.

The approximate pattern is:

broad mathematics $\sim 10^5$,

major mathematical civilization $\sim 10^4$,

strong specialist field $\sim 10^3$,

narrow specialist niche $\sim 10^2$.

This is not a mathematical law.
It is a useful empirical description of how bibliometric ecosystems tend to decompose.
As the field gets narrower, classification uncertainty also increases.

2.15 Why a Few Thousand Is Not “Tiny”

A field of 1,000–3,000 researchers is already a substantial international specialist community.
Likewise,

300

is small relative to major fields but can still support an internationally distributed research niche.

Therefore:

few thousand $\neq$ tiny.

And:

few hundred = small, but not necessarily nonviable.

The classification is about community scale, not mathematical importance.

2.16 What Would Produce Exact Current Numbers?

A genuinely rigorous current census would require a reproducible bibliometric query, ideally using the ZBMATH Open API.

For a field F , publication interval Y , and activity threshold k , define:

$$A(F, Y, k) = \# \{a : P_F(a, Y) \geq k\},$$

where $P_F(a, Y)$ denotes the number of publications by author a assigned to the chosen MSC set defining F during Y .

One could then report:

$$A(F, Y, 1), \quad A(F, Y, 3), \quad A(F, Y, 10).$$

This would distinguish:

broad participants

from

stable active researchers

and

established publishing core

.

The ZBMATH studies explicitly point to the ZBMATH Open API as an infrastructure for more detailed bibliometric analysis [10, 11].

2.17 Final Statistical Picture

The most useful quantitative map is:

Global active mathematical publication ecosystem	$\sim 10^5$ – 2×10^5 ,
Major mathematical civilizations	$\sim 10^4$ – 10^5 ,
Large specialist fields	$\sim \text{several} \times 10^3$,
Narrow specialist fields	$\sim 10^2$ – 10^3 .

For the fields most relevant to the proposed specialization map:

Number Theory	$\sim 10^4$ -scale,
Algebraic Geometry	$\sim 10^3$ – 10^4 ,
Algebraic Topology	$\sim \text{few} \times 10^3$,
Dynamical Systems	$\sim \text{few} \times 10^3$,
Functional/Harmonic Analysis	$\sim \text{few} \times 10^3$,
Representation Theory	$\sim \text{few} \times 10^3$,
Homotopy/Category/Homological fields	$\sim 10^3$ -scale,
Higher-category / HoTT niches	$\sim 10^2$ – 10^3 .

These are *orders of magnitude and ecosystem bands*, not exact headcounts.

Practical interpretation.

If the question is whether a field is roughly a 10^2 , 10^3 , 10^4 , or 10^5 -person ecosystem, the available evidence is useful.

If the question is whether a narrow field contains exactly 2,731 researchers, the available public evidence generally does not justify that precision.

The purpose of this report is the first question.

2.18 Final Statistical Assessment

Claim	Evidence status
Major mathematical ecosystems are five-figure communities	Strongly supported by broad ZBMATH bibliometric clusters.
Number Theory is a large global ecosystem	Strongly supported at the broad ecosystem level.
Topology/Geometry is a large global ecosystem	Strongly supported at the broad ecosystem level.
Dynamics is a substantial specialist ecosystem	Supported by its position within the large DIEq cluster; exact isolated count is inferred.
Algebraic Topology is a few-thousand-scale ecosystem	Reasonable triangulated estimate, not a census.
Homotopy Theory is low-thousands scale	Reasonable specialist estimate, not a direct count.
Category Theory is low-thousands scale	Reasonable specialist estimate, not a direct count.
Higher Category Theory is hundreds-to-low-thousands	Useful community-scale estimate, low confidence.
Higher Algebra is hundreds-to-low-thousands	Useful community-scale estimate, low confidence.
HoTT is hundreds-scale, perhaps approaching low thousands	Community-scale proxy only.
Exact narrow-field researcher counts	Generally not currently defensible without a custom bibliometric query.

Bottom line.

The available evidence is strong enough to distinguish the scale of mathematical communities by orders of magnitude.

It is not strong enough to provide an exact global headcount for most narrow mathematical specialties.

The statistically responsible picture is therefore:

$$10^5 \rightarrow 10^4 \rightarrow 10^3 \rightarrow 10^2$$

with field-specific placement inside that hierarchy and explicit uncertainty.

That is sufficient to answer:

“Roughly how many people are pursuing this area?”

without presenting guesses as census data.

AI-Era Mathematical Specialization

A Deep Audit of a T-Shaped Mathematical Architecture

Evidence, structural reasoning, AI exposure, cognitive compounding, and a proposed vertical–horizontal research strategy

Central thesis.

The durable objective is not to identify mathematics that artificial intelligence cannot do. It is to identify mathematical training whose conceptual benefits compound over time, while increasingly using artificial intelligence to absorb technical and verification-heavy labor.

Pearl Bipin Pulickal

Independent Research Report

September 2026

3. AI-Era Mathematical Specialization

A Deep Audit of a T-Shaped Mathematical Architecture

Evidence, structural reasoning, AI exposure, cognitive compounding, and a proposed vertical–horizontal research strategy

Abstract

This report develops and audits a proposed T-shaped strategy for mathematical specialization in an era of rapidly advancing artificial intelligence (AI). The motivating problem is strategic rather than merely curricular: if AI systems increasingly automate computation, routine symbolic manipulation, proof search, formalization, and even portions of mathematical research, what mathematical training should a human invest in for long-term intellectual and research compounding?

The report deliberately rejects a simplistic “AI-proof mathematics” taxonomy. Current evidence shows that AI capabilities are expanding across algebra, number theory, geometry, topology, formal proof, and research-level problem solving. Systems have reached elite olympiad performance, autonomously resolved nontrivial open conjectures in formal settings, generated and checked large formal proof corpora, and are being designed for longer-horizon mathematical research tasks [16, 18, 20, 21, 22]. These developments make field-level immunity a poor basis for a life-long specialization decision.

Instead, this report distinguishes four things:

- (i) mathematical labor that is increasingly automatable;
- (ii) mathematical work that is highly AI-augmentable;
- (iii) mathematical training that compounds conceptual capacity across multiple domains; and
- (iv) open-ended research activities whose value depends on judgment, problem selection, structural insight, synthesis, and community validation.

On this basis, a provisional architecture is proposed. The principal *vertical stem* is

Algebraic Topology $\longrightarrow$ Homotopy Theory $\longrightarrow$ Homological Algebra,

followed, only after sufficient maturity, by branching toward

Algebraic Geometry, Higher Category Theory, Higher Algebra.

The principal *side lobes* are

Abstract Algebra, Measure Theory / Functional Analysis,
Dynamical Systems / Ergodic Theory, Category Theory.

with secondary expansion into Measure-Theoretic Probability, Number Theory, Representation
and Lie Theory, Harmonic Analysis, and Logic.

The resulting thesis is not that the proposed vertical is inaccessible to AI. Quite the opposite: formalization advances in topology and category theory demonstrate that even highly abstract mathematics can become machine-assisted. The reason for choosing such a vertical is instead *cognitive compounding*: abstraction, invariance, equivalence, local-to-global reasoning, obstruction theory, structural translation, and synthesis can transfer across a wide mathematical ecosystem.

The report concludes that the most robust strategy is:

deep human conceptual training + aggressive AI augmentation

rather than human competition with AI on technical mathematical labor.

Executive Verdict

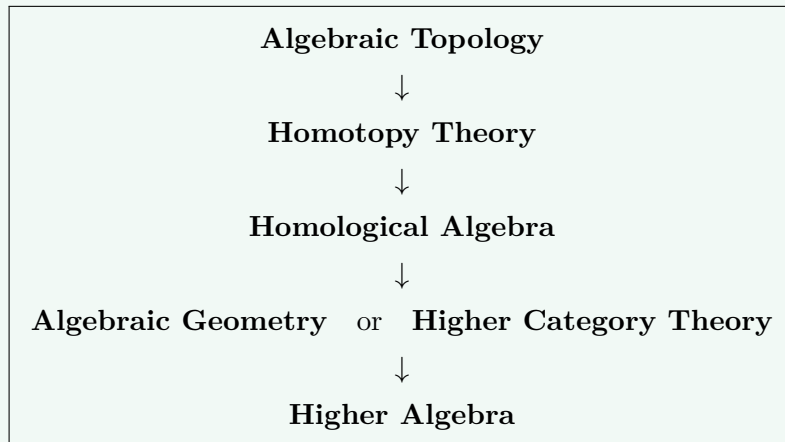
Final recommendation.

After a deliberately adversarial review of the evidence, the proposed T-shaped architecture is retained, but with three corrections.

First, the category “AI-proof” is discarded. No serious long-term forecast can justify it. Second, Higher Category Theory and Higher Algebra are treated as *earned summits*, not as the initial trunk. They should be approached only after substantial grounding in topology, homotopy, algebra, and homological ideas.

Third, the side lobes are pruned into a small number of high-transfer mathematical engines rather than treated as a flat list of subjects.

The recommended architecture is therefore:



with principal lateral engines

Abstract Algebra,
 Measure Theory + Functional Analysis,
 Dynamical Systems + Ergodic Theory,
 Category Theory.

The strategy is therefore not “hide from AI.” It is “become unusually good at deciding what mathematical structure is worth asking AI to explore, formalize, verify, connect, or extend.”

4. Problem Statement

4.1 The strategic problem

A mathematical specialization is a long-horizon investment. The cost of serious mastery is measured not in weeks but in years, and the opportunity cost is substantial. A student cannot maximally specialize in every branch of modern mathematics.

The strategic problem is therefore:

Given rapidly improving AI capabilities, which mathematical architecture maximizes long-term intellectual compounding, research optionality, and productive human–AI collaboration?

This question is deliberately different from:

“Which subject will AI fail to solve?”

The latter is unstable because model capability changes faster than a curriculum can be completed. A specialization chosen because a field appears difficult for AI in 2026 may look very different by 2035.

The former question is more robust because it asks what a human should become *good at* under changing technological conditions.

4.2 The T-shaped model

A T-shaped mathematical education has two components.

The *vertical stem* is deep specialization: a comparatively narrow chain of subjects is studied to a level at which the mathematical language becomes internalized.

The *horizontal bar* contains several neighboring mathematical languages whose combination increases transfer, research mobility, and problem-solving range.

The proposed philosophy can be summarized as

deep vertical + high-transfer side lobes + AI tool layer.

The metaphor used in this report is a *diffraction profile*: deep work in one direction should not remain isolated. It should alter the learner’s capacity to see and organize mathematics across the width.

4.3 A necessary distinction: subject versus task

A central methodological point is that *AI exposure is not a property of a subject alone*.

For example, topology contains at least the following tasks:

1. computing an invariant in a concrete example;
2. checking a proof;
3. searching a theorem library;
4. formalizing a proof;
5. discovering a conjecture computationally;
6. selecting a useful invariant;
7. formulating a research question;
8. synthesizing several theories into a new framework;
9. deciding whether a result is conceptually important;
10. establishing canonical status in a research community.

These tasks can have dramatically different automation profiles.

Thus a useful taxonomy must be written in terms of

task profiles

rather than field labels alone.

5. Methodology and Evidentiary Standard

5.1 Sources

The present report combines four source classes:

1. peer-reviewed or formally archived research concerning AI and mathematical reasoning;
2. current research preprints on mathematical proof search and autonomous research;
3. current documentation and benchmarks for Lean and mathematical formalization;
4. structural judgment about the mathematical transfer properties of the proposed curriculum.

The evidence base includes the 2025–2026 literature on olympiad-level reasoning, AI-driven formal proof search, formal conjecture benchmarks, AI-oriented mathematical research agents, category-theoretic benchmarks, and large-scale formalization [16, 18, 20, 22, 23, 21].

5.2 What counts as strong evidence?

A useful distinction is:

capability evidence $\neq$ community acceptance $\neq$ mathematical understanding.

A machine may generate a formal proof without necessarily providing the most useful conceptual explanation.

Conversely, a system may find a true result while humans still need to establish:

- whether the result is genuinely new;
- whether the formulation is natural;
- whether the proof exposes the correct mechanism;
- whether the result generalizes;
- whether it should become part of the accepted conceptual architecture of the subject.

Terence Tao’s recent discussion of mathematics in the age of AI explicitly separates solving, verification, communication and understanding, community acceptance, and canonicalization, while conditioning on the eventual availability of research-level AI systems [15].

This report therefore avoids treating benchmark scores as complete measures of “mathematical intelligence.”

5.3 The anti-overclaiming rule

The report follows a simple epistemic rule:

A mathematical specialization is recommended only when the positive case survives a plausible attack from the strongest reasonable countercase.

In particular, each proposed subject is tested against the following questions:

1. Is the field already significantly AI-augmented?
2. Is its technical labor likely to become cheaper?
3. Does serious study of it transfer to other fields?
4. Does it train durable structural habits?
5. Does it connect naturally with the proposed vertical?
6. Does it preserve multiple plausible research directions?

6. The AI Landscape Relevant to Mathematics

6.1 Evidence of rapid capability growth

The trajectory from 2024 to 2026 is already sufficient to reject the claim that AI is restricted to elementary mathematics.

In 2024, AlphaProof and AlphaGeometry achieved 28/42 on the IMO, equivalent to silver-medal performance, with several extremely difficult problems solved and formally verified [17, 16].

In 2025, an advanced version of Gemini Deep Think achieved 35/42 on the IMO, a gold-medal-level score, with natural-language proofs produced under the official time limit [18].

The important strategic point is not the competition itself. It is that algebra, number theory, geometry, and combinatorics are no longer reasonable examples of domains in which machine reasoning can safely be regarded as peripheral.

6.2 Research-level formal proof search

More consequentially, AI-driven formal proof search has moved into open research problems.

Tsoukalas et al. report an evaluation in which their strongest agent resolved 9 of 353 open Erdős problems and proved 44 of 492 OEIS conjectures, with applications or deployments spanning combinatorics, optimization, graph theory, algebraic geometry, and quantum optics [20].

This result does *not* imply that AI has mastered mathematics. The success fraction remains a small minority of the tested open problems, and the computational cost is nontrivial.

But strategically it demonstrates something more important:

open-problem mathematics is becoming an AI engineering target.

6.3 AI systems designed for longer-horizon research

The 2026 Aletheia work explicitly frames a transition from competition-level problem solving toward professional mathematical research involving literature navigation, long-horizon proofs, verification, and research-agent workflows. The reported milestones include PhD-level exercises,

open conjecture evaluations, and AI-generated mathematical research artifacts [21].

This evidence should not be read as proof of autonomous mathematicians. Instead, it establishes that the frontier itself is moving toward research workflow automation.

6.4 Formal conjecture infrastructure

The Formal Conjectures benchmark contains 2,615 formalized mathematical problem statements in Lean 4, including 1,029 open research conjectures and 836 solved problems, with an explicit goal of measuring verified discovery rather than merely natural-language plausibility [22].

This is strategically important because it creates a feedback loop:

formal statement $\rightarrow$ machine search $\rightarrow$ formal verification $\rightarrow$ human mathematical evaluation.

Formalization increasingly becomes infrastructure for mathematical AI rather than a niche activity.

6.5 Formal mathematics is scaling

Mathlib now contains more than two million lines of formalized mathematics and supports frontier formalization projects, including the Polynomial Freiman–Ruzsa project and the Liquid Tensor Experiment [27].

The implication is not that formal mathematics is already autonomous. The implication is that mathematical knowledge is becoming increasingly machine-readable, composable, reusable, and searchable.

This is likely to increase the value of people who understand *structure* as well as people who can operate the tools.

7. Why “AI-Proof Mathematics” Is the Wrong Optimization Target

7.1 Field immunity is unstable

Suppose a field F appears difficult for AI at year t . Let $A(F, t)$ denote the fraction of representative tasks that an AI system can perform reliably.

A naive specialization rule would choose

$$F_t^* = \arg \min_F A(F, t).$$

The problem is that the optimizer changes with t :

$$F_{t+1}^* \neq F_t^*$$

may occur frequently when capability improvements are rapid.

A more stable objective is to maximize something like

$$\boxed{\mathcal{V}(F) = C(F) T(F) O(F) L(F),}$$

where:

- $C(F)$ = conceptual compounding;
- $T(F)$ = transfer to neighboring mathematics;
- $O(F)$ = research optionality;
- $L(F)$ = leverage obtainable from AI and mathematical tools.

The formula is not intended as a literal numerical estimator. It is a decision framework.

7.2 The key inversion

As AI becomes more capable, technical automation can actually increase the value of deep conceptual training.

Suppose a human researcher once spent 80 units of effort on:

calculation + proof search + checking

and 20 units on

problem formulation + interpretation.

If AI reduces the first term dramatically, the second term can become the dominant bottleneck.

Hence the strategic shift:

from competing with machines at execution

→ compounding human judgment and structural understanding.

8. The Proposed Vertical Stem

8.1 Algebraic Topology

Algebraic Topology is selected not because it is inaccessible to AI, but because it compresses several unusually transferable mathematical habits.

The central transformations include

space $\longrightarrow$ algebraic invariant,

map $\longrightarrow$ induced structure,

deformation $\longrightarrow$ equivalence.

The learner repeatedly asks:

What information survives deformation?

This is an education in invariance.

It also creates a natural bridge between topology, algebra, geometry, and category-theoretic language.

8.1.1 AI exposure

Topology is already substantially AI-augmented.

Large-scale formalization demonstrates that substantial parts of general topology and combinatorial topology can be expressed in proof assistants. Recent work has formalized substantial pieces of topology and reported very large formal corpora in short periods [25, 26].

Therefore:

Topology is not an AI moat.

But:

Topology remains a powerful human training ground.

That distinction is central.

8.2 Homotopy Theory

Homotopy Theory deepens the topology-to-invariance transition.

The guiding perspective is that strict equality is frequently too rigid, while appropriate notions of equivalence capture the mathematically meaningful structure.

Schematically:

$$X = Y \quad \rightsquigarrow \quad X \simeq Y.$$

This forces the learner to distinguish:

identity, equivalence, deformation, invariant.

Such distinctions have unusually broad conceptual transfer.

8.3 Homological Algebra

Homological Algebra occupies a special place in the proposed stem.

Its central intellectual habit is not merely to construct a structure but to measure the failure of a desired exactness or lifting property.

One repeatedly encounters patterns of the form

$$\text{desired construction} \quad \longrightarrow \quad \text{obstruction} \quad \longrightarrow \quad \text{invariant measuring obstruction}.$$

This makes Homological Algebra particularly valuable as a discipline of controlled abstraction.

It also provides a bridge toward algebraic geometry, representation theory, derived mathematics, and higher categorical structures.

8.4 Algebraic Geometry

Algebraic Geometry is retained as a major branch of the vertical rather than as a compulsory first destination.

Its characteristic pattern

local algebra $\longleftrightarrow$ global geometry

requires the learner to move repeatedly between different mathematical descriptions of the same object.

This is a high-transfer capability.

Its prerequisites are substantial, so making it mandatory too early could create depth at the cost of flexibility.

The proposed rule is therefore:

encounter Algebraic Geometry deeply only when the algebraic and topological foundations are mature enough to make it conceptually productive.

8.5 Higher Category Theory

Higher Category Theory should be regarded as a summit rather than a starting point.

Its mathematical importance is not in dispute here. The issue is sequencing.

An early obsession with maximal abstraction risks confusing familiarity with terminology for structural mastery.

The safer progression is

concrete examples $\rightarrow$ structures $\rightarrow$ maps $\rightarrow$ invariants $\rightarrow$ obstructions $\rightarrow$ higher structures.

This is especially important under modern AI conditions, because formal category theory is itself becoming an explicit benchmark for mathematical reasoning systems. LeanCat introduced a category-theory benchmark designed to stress structural and library-mediated reasoning, demonstrating both the rapid progress and the continuing difficulty of such tasks [23].

8.6 Higher Algebra

Higher Algebra is the natural continuation of the vertical when higher categorical language becomes not merely interesting but useful.

However, the report does not recommend that a learner decide in advance that Higher Algebra *must* be the terminal specialization.

The correct principle is:

Earn the summit rather than worship the summit.

The research trajectory may instead branch into algebraic geometry, homotopy-theoretic dynamics, representation theory, or another interface once genuine mathematical preference emerges.

9. The Primary Side Lobes

9.1 Abstract Algebra

The principal algebraic lobe is

$$\boxed{\text{Groups} \rightarrow \text{Rings} \rightarrow \text{Modules} \rightarrow \text{Fields.}}$$

This is more than a prerequisite sequence.

It is a sequence of increasingly powerful ways of organizing structure.

A recurring abstraction is

object $\rightarrow$ morphism $\rightarrow$ structure-preserving action $\rightarrow$ invariant.

The subject is therefore ideal as horizontal infrastructure.

Assessment: Tier S side lobe.

Extremely high conceptual transfer, strong relevance to the vertical, and extensive AI augmentation for examples, computation, theorem search, formalization, and verification.

9.2 Measure Theory and Functional Analysis

Measure Theory disciplines statements about size, convergence, almost-everywhere behavior, integration, and limiting operations.

Functional Analysis then recasts functions and operators as points in infinite-dimensional spaces.

A useful conceptual progression is:

measure $\rightarrow$ function space $\rightarrow$ operator $\rightarrow$ spectrum / geometry.

The strategic value is not AI resistance.

It is that these subjects provide a rigorous analytical language useful in probability, harmonic analysis, dynamics, PDE, operator theory, and parts of geometry and number theory.

Assessment: Tier S side lobe.

Particularly valuable for broadening a topology/algebra-heavy vertical with high-level analytic thinking.

9.3 Dynamical Systems and Ergodic Theory

Dynamical Systems is arguably the strongest research-facing side lobe in the proposed architecture.

The central object is a map or flow

$$T : X \rightarrow X$$

and the central questions concern long-term behavior:

$$x, T(x), T^2(x), \dots$$

The field naturally asks for:

invariant sets, invariant measures, recurrence, mixing, entropy, stability.

Ergodic Theory then combines dynamics with measure and probability.

This gives the side lobe a remarkable structural profile:

Topology + Analysis + Measure + Probability + Long-term Structure.

AI can be highly effective in numerical exploration, symbolic computation, counterexample generation, simulation, literature navigation, and proof assistance.

But the human research bottleneck can remain:

Which dynamical object matters?

Which invariant captures the phenomenon?

What is the correct theorem?

This makes the field especially well suited to AI augmentation.

Assessment: Tier S side lobe and strong candidate research home.

9.4 Category Theory

Category Theory should be understood primarily as structural infrastructure.

Its central compression is often written informally as

$\text{objects} \rightarrow \text{maps} \rightarrow \text{structure} \rightarrow \text{invariance.}$

The important distinction is that Category Theory should not be used as a “depth amplifier” for arguments that have not first been correctly typed.

A category-theoretic statement cannot repair an ill-defined object.

Thus:

Category Theory is a microscope, not a truth amplifier.

The current formalization landscape reinforces this interpretation. Category theory is increasingly machine-readable and increasingly used as a benchmark for structural reasoning [23, 24].

Assessment: Tier S conceptual side lobe.

10. Secondary Side Lobes

10.1 Measure-Theoretic Probability

Probability is best positioned as a doorway growing naturally from Measure Theory:

Measure Theory $\rightarrow$ Probability $\rightarrow$ Stochastic Processes $\rightarrow$ Ergodic / Random Dynamics.

This avoids treating probability as a detached computational subject.

The mathematical formulation is especially useful when the goal is structural understanding rather than formula memorization.

10.2 Number Theory

Number Theory is a high-value optional lobe.

Its range includes:

elementary arithmetic $\rightarrow$ algebraic number theory
 $\rightarrow$ analytic number theory $\rightarrow$ arithmetic geometry.

It is also a natural testing ground for computational experiments and conjecture generation.

AI capability is rising rapidly in this ecosystem, so it should not be selected because of presumed resistance. Instead, it is attractive because it provides a rich interface between algebra, analysis, geometry, computation, and dynamics.

10.3 Representation Theory and Lie Theory

Representation Theory teaches a particularly transferable pattern:

complicated object $\rightarrow$ action on simpler space.

The study of symmetry through representations creates connections among algebra, geometry, analysis, combinatorics, and physics.

Lie Theory adds the continuous analogue of symmetry:

group $\longleftrightarrow$ infinitesimal structure.

These are therefore excellent second-wave investments once the algebraic base is solid.

10.4 Harmonic Analysis

Harmonic Analysis provides an especially useful bridge:

analysis $\longleftrightarrow$ dynamics $\longleftrightarrow$ number theory $\longleftrightarrow$ PDE.

It is therefore a high-value advanced side lobe, although not necessarily an initial horizontal pillar.

10.5 Logic

Logic provides epistemic discipline:

statement $\neq$ intuition $\neq$ proof $\neq$ model.

That separation is valuable in any research field.

At the same time, logic and formal reasoning are highly AI-compatible. Consequently, Logic is recommended for the precision it instills, not for supposed automation resistance.

11. Comparative Assessment

11.1 Decision dimensions

The following qualitative scores use a five-level ordinal scale.

- **Conceptual compounding:** how strongly mastery changes the learner’s future mathematical representations.
- **Transfer:** how many neighboring fields receive substantial benefit.
- **AI:** how much the field can benefit from computation, proof search, formalization, and AI assistance.
- **Research optionality:** number and diversity of plausible future research interfaces.
- **Prerequisite burden:** amount of preparation required before the subject becomes deeply productive.

Field / cluster	Comp.	Trans.	AI	Opt.	Pre.
Algebraic Topology	5	5	4	5	4
Homotopy Theory	5	5	4	5	5
Homological Algebra	5	5	4	5	5
Algebraic Geometry	5	5	4	5	5
Higher Category Theory	5	4	5	4	5
Higher Algebra	5	5	5	5	5
Abstract Algebra	5	5	5	5	3
Measure Theory	5	5	5	5	3
Functional Analysis	5	5	5	5	4
Dynamical Systems	5	5	5	5	4
Ergodic Theory	5	5	5	5	5
Category Theory	5	5	5	5	4
Probability	4	5	5	5	4
Number Theory	5	5	5	5	3
Representation Theory	5	5	5	5	4
Lie Theory	4	4	5	4	4
Harmonic Analysis	5	5	5	5	4
Logic	5	5	5	4	3

Interpretation. The table is not a claim that the fields can be objectively scored in a universal way. It encodes the strategic argument of this report and is intentionally qualitative.

11.2 AI exposure: a task-based taxonomy

A more useful classification is:

Task class	Likely AI trajectory	Human strategic response
Routine symbolic manipulation	High automation	Offload aggressively
Numerical experimentation	High automation / augmentation	Use as exploratory machinery
Formal proof checking	Very high automation	Treat as infrastructure
Proof search in bounded settings	Rapidly improving	Use AI, then inspect structurally
Counterexample search	Rapidly improving	Make it part of the audit loop
Literature retrieval and synthesis	High augmentation	Use AI to increase breadth
Problem selection	Less standardized	Cultivate taste and judgment
Definition design	Partly augmentable	Retain human ownership of meaning
Invariant selection	Partly augmentable	Develop structural intuition
Cross-domain synthesis	Rapidly improving but difficult	Use AI for alternatives; retain judgment
Canonicalization / meaning	Slow and community-dependent	Invest heavily in understanding
Research taste	No reliable benchmark	Cultivate through years of mathematics

12. Why the Proposed Vertical Is a Cognitive Investment

12.1 The compounding thesis

A subject is especially valuable as a vertical when mastering it changes the way later mathematics is represented.

Let M_0 denote a learner's initial mathematical representation system. After deep study of a subject F , write

$$M_1 = \Phi_F(M_0),$$

where Φ_F is the transformation induced by sustained study.

A “high-compounding” subject is one for which

$$\Phi_F$$

improves performance not only inside F but across many neighboring mathematical domains.

The claim of this report is that Algebraic Topology, Homotopy Theory and Homological Algebra have unusually high potential for such cross-domain transformation.

The reason is not mysticism. It is structural repetition.

The learner repeatedly encounters:

objects, maps, equivalences, invariants,
obstructions, functoriality, local-to-global transitions.

12.2 Why this matters under AI

Suppose AI gradually reduces the cost of obtaining

calculation, proof sketch, formal certificate, example generation.

Then the human bottleneck may increasingly lie in choosing:

which object?

which representation?

which invariant?

which conjecture?

which generalization?

which obstruction?

The proposed vertical is therefore attractive because it repeatedly trains precisely these kinds of representational choices.

13. The Dynamics / Ergodic Research Doorway

13.1 Why Dynamics is unusually compatible with the architecture

Dynamical Systems provides a research-facing doorway through which the algebraic/topological and analytical/probabilistic sides can interact.

A general dynamical system may be represented as

$$(X, T)$$

or

$$(X, \varphi^t).$$

The questions include:

What is invariant?

What is recurrent?

What asymptotic structures emerge?

Which perturbations preserve the phenomenon?

These questions naturally connect to topology, measure, probability, functional analysis, harmonic analysis, number theory, and geometry.

13.2 Complex Dynamics

Complex Dynamics adds the structure of holomorphic maps.

A prototype is iteration

$$z_{n+1} = f(z_n),$$

where the long-run behavior depends on the geometry of the map.

The conceptual combination

complex analysis + dynamics + geometry

creates a particularly rich research environment.

13.3 Arithmetic Dynamics

Arithmetic Dynamics instead studies iteration while retaining arithmetic structure:

$$x_{n+1} = f(x_n), \quad x_n \in K$$

for suitable number-theoretic spaces K .

This creates a powerful interface:

dynamics + number theory.

For a broad researcher, this is strategically attractive because the same dynamical intuition can migrate between analytic, geometric and arithmetic settings.

13.4 Research-home versus cognitive-home

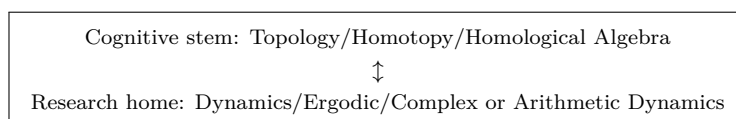
The report distinguishes two different roles.

The *cognitive home* is the vertical stem whose primary function is deep abstraction and intellectual compounding.

The *research home* is the region where one actually wants to formulate problems and publish.

These need not be identical.

A mathematically coherent configuration is therefore:



This is one of the strongest versions of the T-shaped strategy.

14. Failure Modes of the Strategy

14.1 Failure mode 1: chasing AI-resistance

The first and most important failure mode is optimizing for a property that may disappear.

AI-resistant today $\nrightarrow$ AI-resistant later.

The remedy is to optimize for:

compounding + transfer + AI leverage.

14.2 Failure mode 2: maximal abstraction too early

Higher Category Theory and Higher Algebra are intellectually attractive. That does not mean they should be started at maximum intensity.

The failure mode is:

abstract vocabulary $\rightarrow$ illusion of depth.

The remedy is:

examples $\rightarrow$ structures $\rightarrow$ maps $\rightarrow$ invariants $\rightarrow$ higher abstraction.

14.3 Failure mode 3: excessive breadth

A T-shaped curriculum can collapse into a horizontal bar with no stem.

The warning sign is:

many books + many playlists + many subjects – deep mastery.

A useful T must have an unmistakable vertical.

14.4 Failure mode 4: using AI as a substitute for understanding

An AI-generated proof can produce:

P .

That does not imply the learner possesses:

Understanding(P).

The distinction matters because future mathematical value may concentrate in people who can interpret, generalize, critique, and use machine-generated results rather than merely reproduce them.

14.5 Failure mode 5: confusing formal correctness with significance

Formal verification establishes something very important:

the formal statement follows from the formal assumptions.

It does not automatically establish:

the theorem is conceptually natural,

the theorem is important,

or

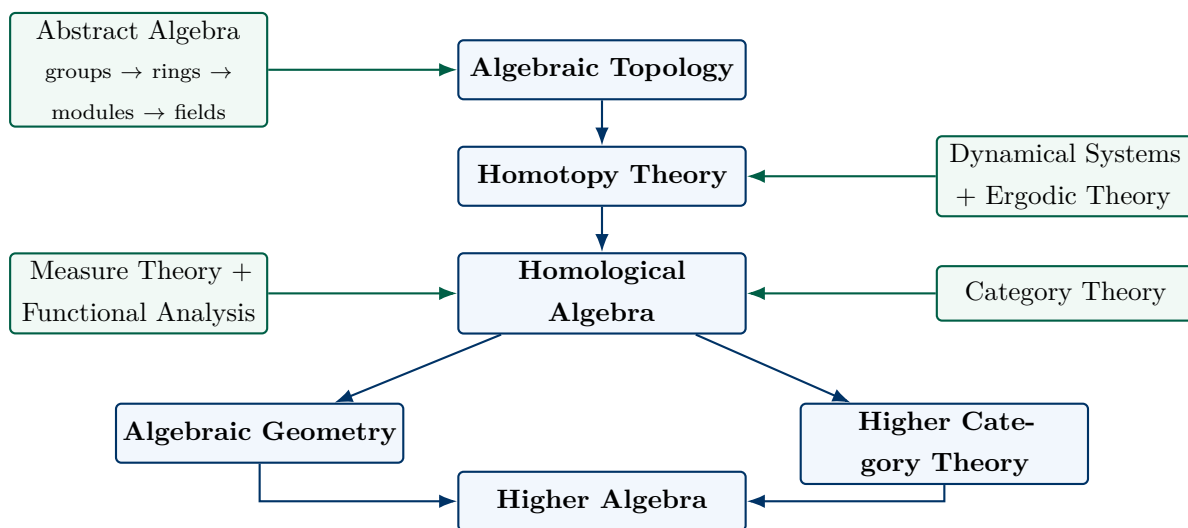
the theorem reveals the right explanatory mechanism.

This distinction is especially relevant as formal mathematical corpora grow.

15. The Final Mathematical Architecture

15.1 Core architecture

The recommended architecture can be displayed as:



15.2 Secondary expansion

Once the core architecture is stable, the following subjects can be added as optional expansion directions:

Measure-Theoretic Probability,
Number Theory,
Representation Theory,
Lie Theory,
Harmonic Analysis,
Logic.

These are not a checklist requiring equal mastery.

They are branches from which research identity can later emerge.

15.3 Research-interface map

The architecture permits several high-value interfaces:

Topology/Homotopy $\leftrightarrow$ Dynamics,
Dynamics $\leftrightarrow$ Ergodic Theory,
Ergodic Theory $\leftrightarrow$ Probability,
Functional Analysis $\leftrightarrow$ Dynamics,
Number Theory $\leftrightarrow$ Arithmetic Dynamics,
Complex Analysis $\leftrightarrow$ Complex Dynamics,
Algebra $\leftrightarrow$ Representation Theory,
Homological Algebra $\leftrightarrow$ Algebraic Geometry,
Homotopy $\leftrightarrow$ Higher Category Theory.

This network is an advantage: the learner is not trapped in a single research niche.

16. A Practical Learning Sequence

16.1 Stage I: Build the structural base

The first layer should emphasize:

Algebra + Analysis + Topology.

The goal is not maximal breadth. The goal is to become fluent in the language required by the vertical.

A reasonable base includes:

- groups, rings, modules and fields;
- point-set topology;
- real analysis and measure;
- functional analysis;
- elementary category theory.

16.2 Stage II: Deepen topology

Then:

Algebraic Topology $\rightarrow$ Homotopy.

At this point the student should begin solving substantial exercises, rewriting proofs independently, constructing examples, searching for counterexamples, and using AI only as an accelerator and adversarial checker.

16.3 Stage III: Introduce homological thinking

Then:

Homotopy $\rightarrow$ Homological Algebra.

The goal is to internalize:

obstruction, exactness, derived information.

16.4 Stage IV: Open the research interface

In parallel:

Dynamics + Ergodic Theory.

This stage tests whether the vertical produces actual research energy rather than merely abstract admiration.

16.5 Stage V: Choose the summit

Only after substantial experience should the learner choose among:

Algebraic Geometry, Higher Category Theory, Higher Algebra, Dynamics/Homotopy interfaces.
--

or another field revealed by genuine mathematical preference.

17. AI as the Mathematical Tool Layer

17.1 The proposed division of labor

The strategy advocates neither rejection of AI nor passive dependence on it.

Instead:

Human should emphasize	AI should increasingly absorb
Problem selection	Routine theorem search
Definition design	Symbolic manipulation
Structural intuition	Numerical experiments
Invariant selection	Counterexample search
Conceptual synthesis	Formalization assistance
Research taste	Proof search
Interpretation	Code generation
Generalization	Literature retrieval
Mathematical judgment	Routine verification
Canonicalization	Exploratory computation

The division is not absolute. A good research workflow repeatedly crosses the boundary.

17.2 The ideal loop

A robust human–AI mathematical loop is:

Human–AI research loop

Human question → AI exploration
→ formal / computational checking
→ human interpretation → revised question.

The loop is superior to both extremes:

human-only

and

AI-only.

17.3 Adversarial verification

The most important operational use of AI may be neither generation nor exposition but *attack*.

For each serious mathematical idea, ask an AI system to search for:

1. the smallest nontrivial case;
2. the degenerate case;
3. the boundary case;
4. asymmetry;
5. order dependence;
6. hidden type mismatches;
7. missing hypotheses;
8. counterexamples;
9. failure of functoriality;
10. non-equivalence of apparently analogous constructions.

This turns AI into an adversarial auditor rather than a truth oracle.

18. Why This Architecture Is Particularly Coherent

18.1 Multiple representations of the same phenomenon

A strong mathematical education permits movement between:

algebraic, topological, analytic, probabilistic, dynamical
representations.

This is a form of mathematical compression.

A phenomenon may be understood through:

equations,

operators,

spaces,

symmetries,

invariants,

or

dynamics.

The T-shaped architecture deliberately maximizes the number of bridges among these representations.

18.2 A graph-theoretic interpretation

Let the mathematical domains be vertices of a graph G and let an edge denote a strong conceptual transfer.

The value of a curriculum need not be proportional to the number of vertices studied.

A more useful heuristic is to maximize:

depth of selected vertices $\times$ connectivity of the induced subgraph.

The proposed vertical provides depth.

The proposed side lobes provide connectivity.

The research interfaces provide bridges.

18.3 Why a small number of side lobes is better

Suppose each side lobe requires annual maintenance cost c_i . If available time is B , then:

$$\sum_i c_i \leq B.$$

As the number of side lobes increases, depth tends to decrease unless the learner has an unusually large time budget.

Hence:

few high-transfer lobes > many shallow lobes.

This is one reason the final architecture emphasizes four primary side lobes rather than treating every attractive field equally.

19. Robustness Analysis

19.1 Counterfactual 1: AI improves much faster

Suppose AI becomes substantially better at:

proof + formalization + conjecture generation.

Then the strategy becomes *more* valuable in its AI-layer component.

The vertical remains valuable because its purpose was never technical exclusivity.

19.2 Counterfactual 2: AI progress slows

Suppose AI improvement becomes much slower.

The strategy remains viable because it is already a mathematically coherent human curriculum.

Thus the strategy does not depend on any single AI forecast.

19.3 Counterfactual 3: Higher mathematics becomes highly formalized

Suppose most advanced mathematics becomes formally represented.

Then:

formal literacy

becomes more important, not less.

A researcher who understands the mathematics behind the formal structures can exploit the resulting infrastructure more effectively.

19.4 Counterfactual 4: autonomous research becomes genuinely strong

Suppose research agents eventually become capable of solving many well-posed research problems autonomously.

Then the bottleneck may move further toward:

problem selection + research taste + meaning + canonicalization + scientific direction.

This report's strategy is designed for precisely that possibility.

20. Limitations

No report of this type can provide certainty.

First, AI capability is changing quickly. A classification that is reasonable in September 2026 may be stale within a few years.

Second, the available benchmark evidence is not a representative sample of all mathematical work. Many systems are evaluated on selected problems, formal corpora, or highly structured tasks.

Third, capability demonstrations are not identical to autonomous research careers. The latter requires sustained problem selection, novelty assessment, communication, collaboration, and community validation.

Fourth, “cognitive compounding” is a useful strategic concept but is not a directly measurable scalar.

Fifth, the proposed ordering is not a universal prescription. A different learner may rationally choose PDE, probability, combinatorics, logic, geometry, or number theory as the vertical.

Sixth, personal mathematical preference matters enormously. No strategic architecture can compensate indefinitely for lack of genuine interest.

Important.

The recommendations in this report should be interpreted as an evidence-based decision framework, not as a prediction that any specific mathematical field will remain human-dominated or AI-resistant.

21. Final Audit

21.1 Question 1: Is the vertical mathematically coherent?

Yes.

The chain

Algebraic Topology $\rightarrow$ Homotopy $\rightarrow$ Homological Algebra

is structurally coherent and leads naturally toward several advanced interfaces.

21.2 Question 2: Is it AI-proof?

No.

And it should not be advertised that way.

Large-scale topology formalization, formal category theory, and research-level proof search demonstrate that advanced abstraction is increasingly machine-assisted [25, 23, 20].

21.3 Question 3: Is it still strategically attractive?

Yes.

Its attractiveness comes from:

structural depth + transfer + research optionality + AI leverage.

21.4 Question 4: Are the side lobes coherent?

Yes, especially:

Abstract Algebra,

Measure Theory + Functional Analysis,

Dynamics + Ergodic Theory,

Category Theory.

These four form a strong horizontal platform.

21.5 Question 5: Is Dynamics a plausible research home?

Yes.

Among the proposed side lobes it has one of the strongest intersections with the rest of the architecture:

topology + measure + analysis + probability + geometry.

Complex and Arithmetic Dynamics then open additional bridges to complex analysis and number theory.

21.6 Question 6: Should Higher Category Theory be studied immediately?

No.

It should be approached after sufficient lower-level structural experience.

21.7 Question 7: Should AI be treated as the opponent?

No.

The best strategy is:

human conceptual depth + machine technical leverage.
--

21.8 Question 8: What survives if the AI forecast is wrong?

Almost everything.

The curriculum remains a powerful mathematical education even if AI progress slows dramatically.

That is an essential robustness property.

22. Conclusion

The final conclusion of this report can be compressed into one sentence:

Do not choose mathematics by searching for a permanent machine-free zone; choose a mathematical architecture in which deep human understanding and increasingly powerful mathematical machines reinforce one another.

The proposed T therefore has three layers.

VERTICAL

Algebraic Topology $\rightarrow$ Homotopy Theory $\rightarrow$ Homological Algebra $\rightarrow$ advanced branch

HORIZONTAL

Abstract Algebra + Measure/Functional Analysis + Dynamics/Ergodic Theory + Category Theory

TOOL LAYER

AI + formal verification + computation + literature systems.

The central strategic insight is therefore not that the vertical must remain beyond AI.

It is that the vertical should make the learner *better at seeing mathematics as structure*.

That structural vision can then radiate laterally:

objects $\rightarrow$ maps $\rightarrow$ structure $\rightarrow$ invariance $\rightarrow$ obstruction $\rightarrow$ higher structure.

Dynamics supplies the experimental and research-facing doorway:

structure $\rightarrow$ evolution $\rightarrow$ invariant behaviour $\rightarrow$ long-term organization.

AI supplies leverage:

search + computation + proof assistance + verification + formalization.

The resulting philosophy is neither defensive nor utopian.

It is simply:

Go deep enough that your mathematical representations change.

Go broad enough that those representations transfer.

Use machines aggressively for everything that should become cheap.

Reserve human effort for structure, meaning, judgment, and discovery.

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A. Compact Decision Matrix

Field	Primary role	AI stance	Final recommendation
Algebraic Topology	Vertical foundation	Strongly augmentable	Core
Homotopy Theory	Vertical deepening	Strongly augmentable	Core
Homological Algebra	Vertical bridge	Strongly augmentable	Core
Algebraic Geometry	Advanced branch	Strongly augmentable	Earned branch
Higher Category Theory	Summit	Strongly augmentable	Earned summit
Higher Algebra	Summit	Strongly augmentable	Earned summit
Abstract Algebra	Horizontal infrastructure	Excellent AI leverage	Core lobe
Measure Theory	Analytical infrastructure	Excellent AI leverage	Core lobe
Functional Analysis	Analytical infrastructure	Excellent AI leverage	Core lobe
Dynamical Systems	Research-facing lobe	Excellent AI leverage	Core lobe
Ergodic Theory	Research-facing lobe	Excellent AI leverage	Core lobe
Category Theory	Structural language	Excellent AI leverage	Core lobe
Probability	Bridge to stochastic dynamics	Excellent AI leverage	Secondary
Number Theory	Arithmetic expansion	Excellent AI leverage	Secondary
Representation Theory	Symmetry interface	Excellent AI leverage	Secondary
Lie Theory	Continuous symmetry	Excellent AI leverage	Secondary
Harmonic Analysis	Analysis interface	Excellent AI leverage	Secondary
Logic	Precision / foundations	Excellent AI leverage	Secondary

B. Research Questions for External Review

The report is intended to be discussable rather than merely declarative. A useful external audit should therefore address:

1. Does Algebraic Topology genuinely have unusually high cross-field cognitive transfer, or is that merely aesthetic preference?
2. Is Homological Algebra correctly positioned as the bridge between topology and higher structures?
3. Should Algebraic Geometry outrank Higher Category Theory as the natural next branch?
4. Is Dynamics/Ergodic Theory the strongest research-facing side lobe, or is another interface superior?
5. Are Measure Theory and Functional Analysis sufficiently central in a topology-first architecture?
6. Which important fields have been omitted?
7. Which assumptions about AI capabilities are too optimistic?
8. Which assumptions about human comparative advantage are too optimistic?
9. Does the curriculum remain valuable if AI capability increases dramatically over the next decade?
10. Does the curriculum remain valuable if AI capability advances much more slowly than expected?

These questions are intentionally adversarial. The architecture should be revised if competent external criticism reveals a better structure.

C. A Minimal Operational Version

For practical use, the entire report can be compressed to the following:

VERTICAL

Algebraic Topology $\rightarrow$ Homotopy $\rightarrow$ Homological Algebra

then choose: Algebraic Geometry / Higher Category Theory /

Higher Algebra

SIDE LOBES

Abstract Algebra + Measure + Functional Analysis + Dynamics + Ergodic + Category Theory

SECONDARY

Probability, Number Theory, Representation/Lie Theory,

Harmonic Analysis, Logic

TOOL LAYER

AI + formalization + computation + verification

The optimization target is:

conceptual compounding + transfer + research optionality + AI leverage.

Not:

AI-proofness.

D. Author's Current Candidate T-Shaped Mathematics Career

This appendix records the author's current personal candidate architecture. It is intentionally separated from the evidence-based and theory-driven main body of the manuscript. Its role is not to turn a personal preference into a universal prescription, but to make the concrete specialization that motivates the broader framework explicit enough to be examined, revised, and tested over time.

The candidate is a *T-shaped mathematical career with structured side lobes*: a deep vertical stem, a set of high-transfer mathematical side lobes, and a smaller set of interface or doorway subjects through which the researcher can enter multiple research ecosystems.

Personal candidate, not universal recommendation.

The architecture below is the author's current preferred configuration. It is compatible with the main theory's distinction between a cognitive stem, secondary competence, and a research home. The ordering is a proposed developmental architecture rather than a claim that every item is a strict formal prerequisite for the next.

D.1 The vertical stem

The current preferred deep stem is:

Algebraic Topology
→ Homotopy Theory
→ Homotopy Type Theory
→ Homological Algebra
→ Algebraic Geometry
→ Higher Category Theory
→ Higher Algebra.

The intended intellectual progression is from topological structure and invariance, through homotopy and formal structural language, toward homological methods, algebraic geometry, higher categorical organization, and higher algebra. Homotopy

Type Theory is retained as part of the preferred stem because the author regards the interaction among homotopy, type-theoretic foundations, formal reasoning, and higher structures as especially relevant to the long-term architecture. The sequence should not be interpreted as asserting that each subject is a necessary formal prerequisite for the next; it is a proposed order of intellectual development.

D.2 Top side lobes

The preferred high-transfer lateral regions are:

- Dynamical Systems;
- Ergodic Theory;
- Complex Dynamics / Arithmetic Dynamics;

- Number Theory;
- Harmonic Analysis;
- the algebraic chain of Groups $\rightarrow$ Rings $\rightarrow$ Modules $\rightarrow$ Fields;
- Group Theory, Representation Theory, Lie Theory, and related algebraic areas;
- Measure Theory and Functional Analysis;
- Measure-Theoretic Probability;
- the Dynamics/Ergodic research doorway;
- Logic / Category Theory;
- the structural sequence *objects* $\rightarrow$ *maps* $\rightarrow$ *structure* $\rightarrow$ *invariance*.

These are not intended to be treated as independent verticals of equal depth. They are candidate regions in which the representations developed by the vertical stem may radiate into other parts of mathematics. In the language of the main theory, these are candidate side lobes whose strength is expected to depend on structural resonance rather than on surface similarity alone.

D.3 The Dynamics/Ergodic doorway

Among the side lobes, the author currently regards Dynamics and Ergodic Theory as an especially important research-facing doorway. The reason is structural rather than defensive: the area creates an interface among topology, measure, probability, analysis, geometry, and long-term evolution. The research-home concept in the main manuscript is designed to allow exactly this kind of configuration, where the cognitive stem and the eventual publication identity need not coincide.

D.4 Structural algebraic side lobe

The algebraic infrastructure is currently conceived as

$$\text{Groups} \rightarrow \text{Rings} \rightarrow \text{Modules} \rightarrow \text{Fields},$$

with extensions into Group Theory, Representation Theory, Lie Theory, and related algebraic areas. The purpose is not to collect algebraic subjects as labels but to develop a reusable language of objects, morphisms, actions, representations, symmetry, and invariance.

D.5 Analytic and probabilistic side lobes

The analytic side of the architecture is centered on Measure Theory and Functional Analysis, with Measure-Theoretic Probability as a bridge into stochastic and ergodic structures. Harmonic Analysis is treated as a further interface connecting analysis, dynamics, number theory, and PDE.

D.6 Logic and Category Theory as structural infrastructure

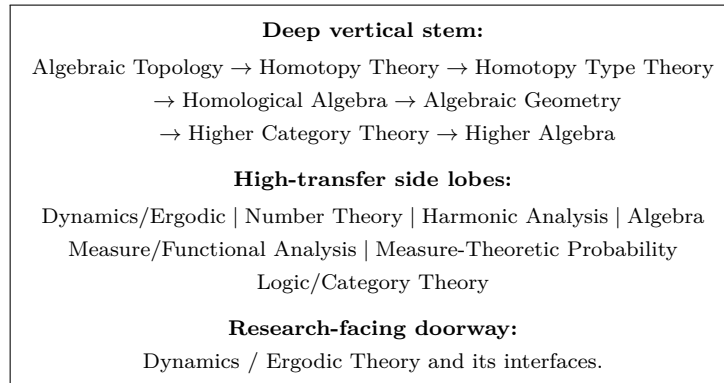
Logic and Category Theory are positioned somewhat differently from ordinary side lobes. They are viewed as structural languages that clarify statements, maps, compositionality, invariance, and the relationships among mathematical theories.

The guiding compression is

$\text{objects} \rightarrow \text{maps} \rightarrow \text{structure} \rightarrow \text{invariance}.$
--

D.7 Current personal synthesis

The resulting personal configuration can be summarized as:



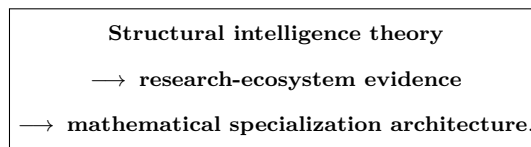
This appendix is deliberately personal. The main manuscript asks whether the architecture is coherent, evidence-informed, and robust under criticism. This appendix records what the author would currently choose as the concrete realization of that architecture, subject to future mathematical experience and external criticism.

Integrated Closing Note

This integrated manuscript deliberately retains the three original bodies of work rather than compressing them into a summary. Chapter 1 establishes a mathematical hypothesis about the geometry of expertise and structural transfer.

Chapter 2 supplies the quantitative research-ecosystem context needed to understand the scale of the mathematical communities into which specialization occurs. Chapters 3 onward apply the theory and ecosystem evidence to a concrete mathematical specialization architecture under rapidly changing AI capabilities.

The resulting logical sequence is therefore:



The final personal appendix records one concrete candidate realization of that architecture. It is intentionally presented as a living research and career hypothesis rather than as a theorem about what every mathematician should study. The architecture may change as mathematical knowledge, empirical evidence, and AI capabilities change.

The manuscript is strongest when each layer remains distinguishable: the theory can be tested; the ecosystem estimates can be improved; and the specialization architecture can be attacked, revised, or replaced. Their convergence is the research program.